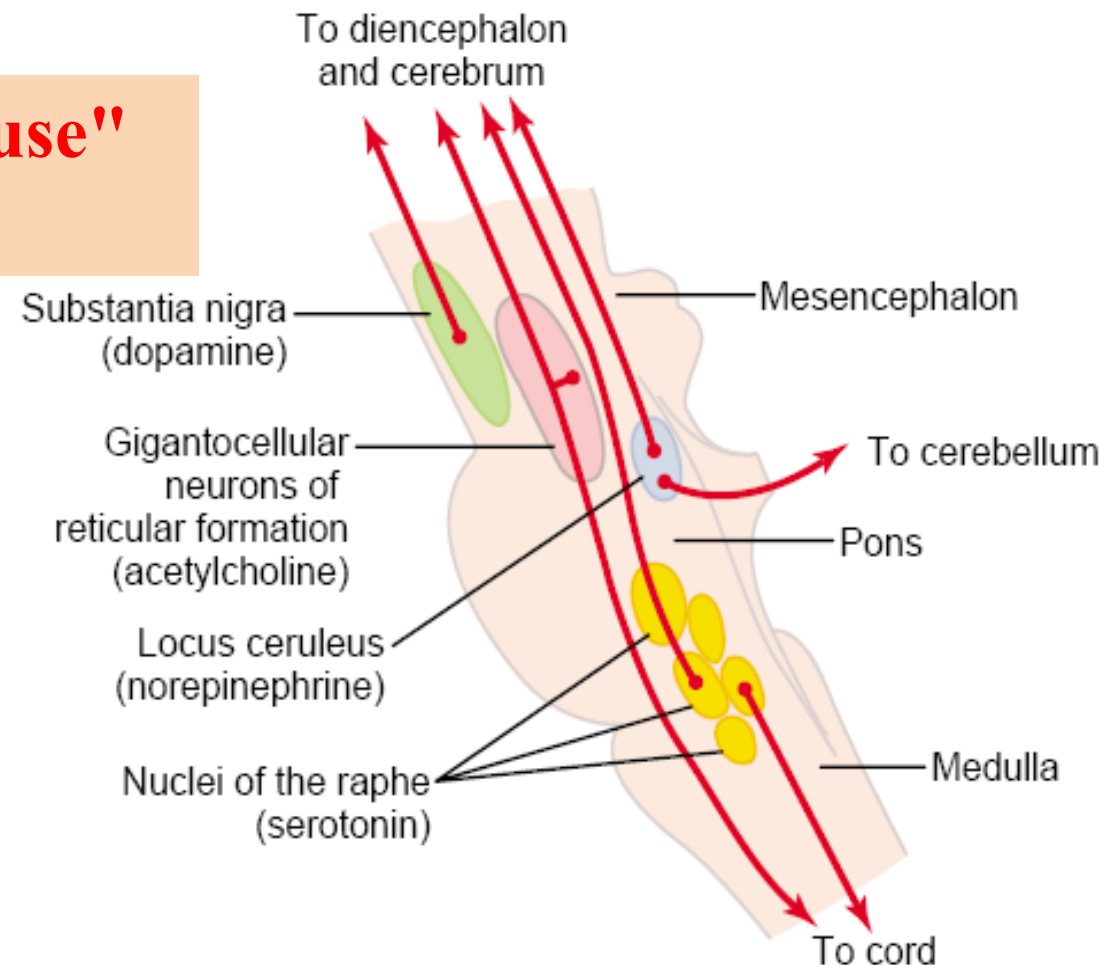


Neurohormonal Control of Brain Activity

- Activation of the Brain in addition to RAS system – **lasts for min.-hrs**
- One system arises in the brainstem RF and sends projections to the cortex via non-specific thalamic nuclei (*Non-specific = diffuse nuclei*).
- Other system consists of brainstem nuclei which release neurohormones to various brain regions via ascending projections.

The "non-specific" or "diffuse" thalamocortical system.

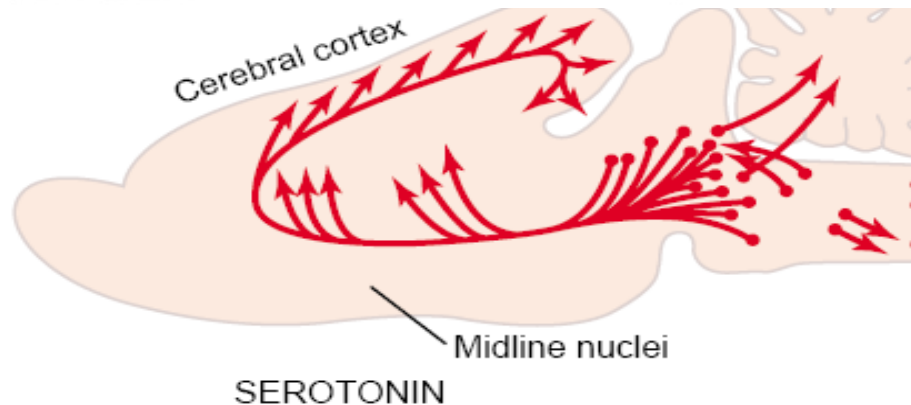
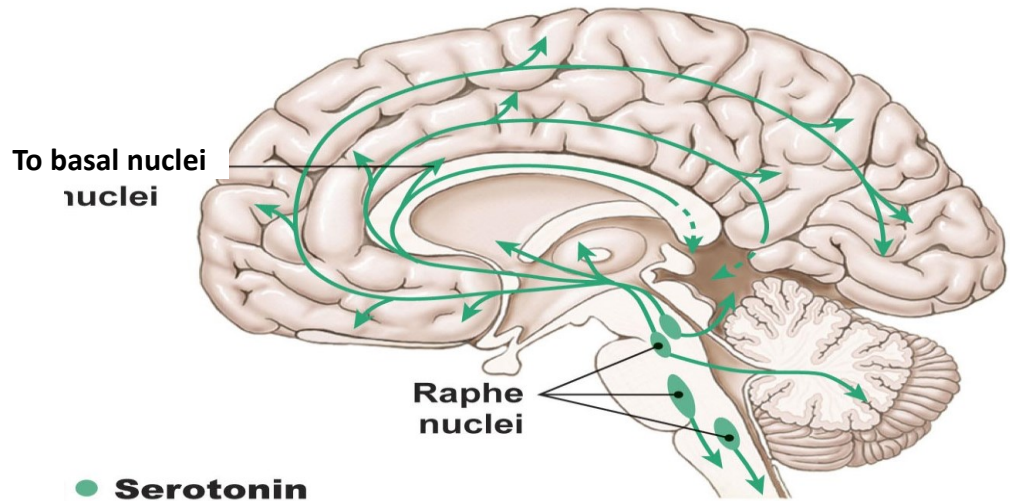
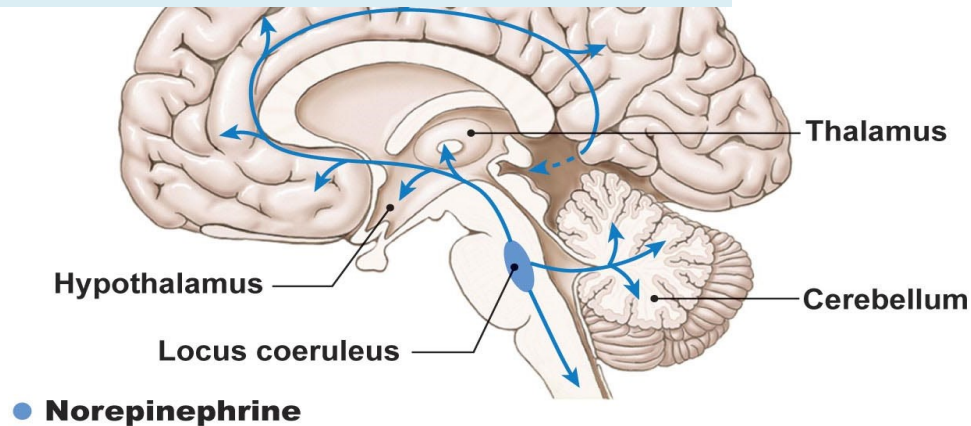
- Three systems
 - **Norepinephrine**
 - **Dopamine and**
 - **Serotonin**



The three/four? systems: Monoaminergic systems

Over activity = anxiety

Underactivity = depression



1. The Norepinephrine System

- Arise as the locus ceruleus (LC)
- Send projections to all parts of the CNS.
- Associated with REM sleep
- Set mood (attentivness) and affect (emotion; e.g., euphoria)
- Amphetamines and cocaine boosting

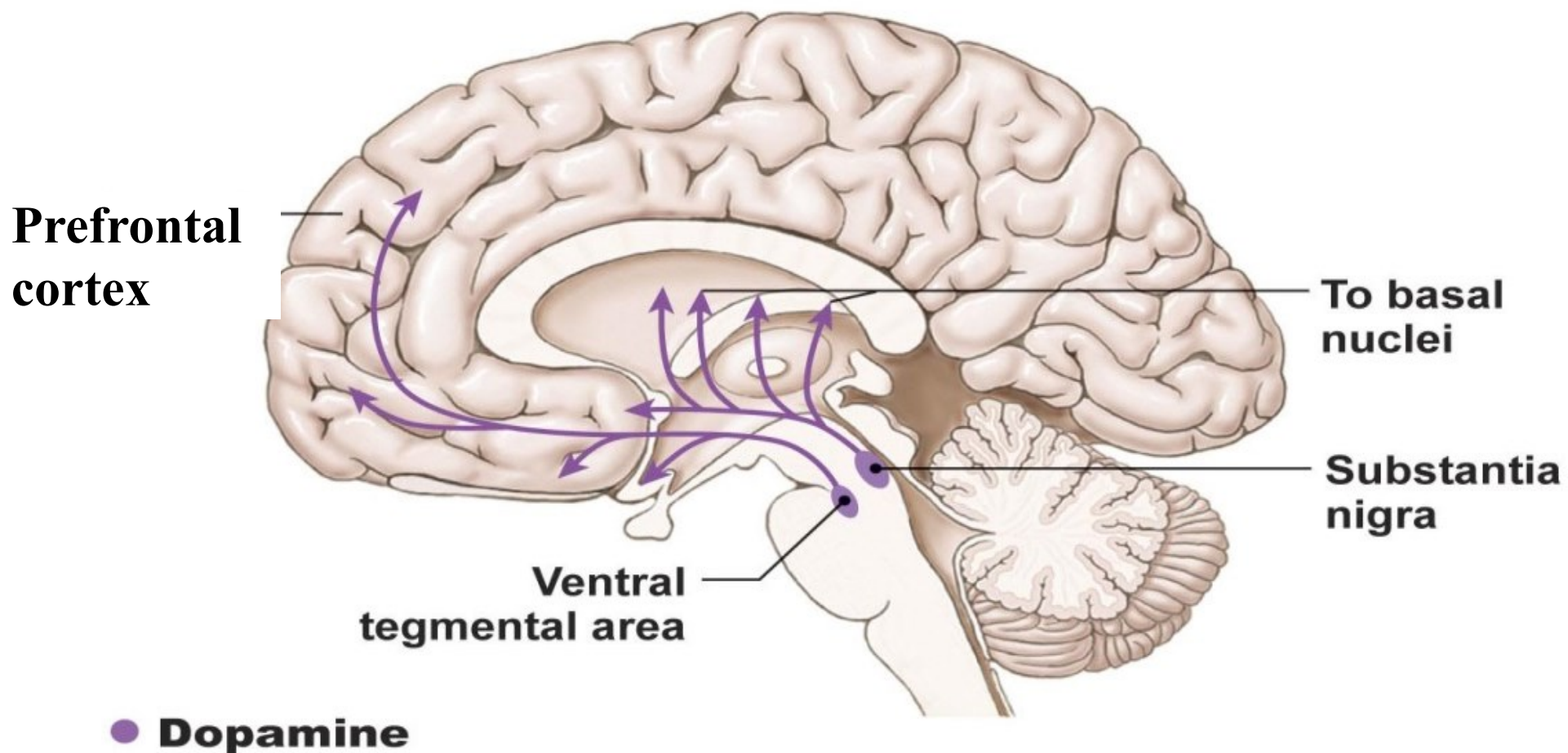
2. The Serotonin System

- Primarily located on raphe nuclei.
- Mood, memory, sleep, & cognitive skills.
- Projections to diencephalon and cortex and also to spinal cord.
 - *"sleep centers"*
- Projections to the spinal cord inhibit transmission of pain signals over the spinothalamic tracts
 - *brains analgesic system.*

3. The Dopamine System

- SN = project into the basal ganglia (putamen and caudate nuclei)
 - *Inhibitory effects are on controlling factor in the voluntary motor system.*
- VTA= project to various areas of the forebrain including the nucleus accumbens, hypothalamus, septal area, amygdala, and the cerebral cortex.
 - Plays a role in *motivation , drive and cognition*

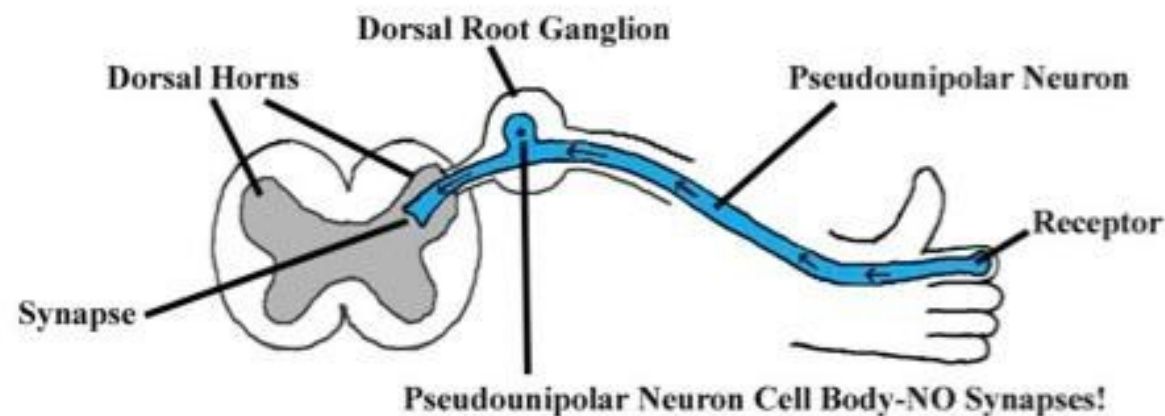
...cocaine inhibits dopamine reuptake, and amphetamine promotes dopamine release and inhibits its reuptake = both **stimulate brain's reward system.**



SYSTEM (NEUROMODULATOR)	SITE WHERE NEURONS ORIGINATE	STRUCTURES THAT NEURONS INNERVATE	FUNCTIONS MODULATED BY THE SYSTEM
Noradrenergic (norepinephrine)	Locus coeruleus of the pons	Cerebral cortex, thalamus, hypothalamus, olfactory bulb, cerebellum, midbrain, spinal cord	Attention, arousal, sleep-wake cycles, learning, memory, anxiety, pain, and mood
Serotonergic (serotonin)	Raphe nuclei along brain stem midline	Lower nuclei project to spinal cord	Pain, locomotion
		Upper nuclei project to most of brain	Sleep-wake cycle; mood and emotional behaviors, such as aggression and depression
Dopaminergic (dopamine)	Substantia nigra in midbrain	Cortex	Motor control
	Ventral tegmentum in midbrain	Cortex and parts of limbic system	"Reward" centers linked to addictive behaviors
Cholinergic (acetylcholine)	Base of cerebrum; pons and midbrain	Cerebrum, hippocampus, thalamus	Sleep-wake cycles, arousal, learning, memory, sensory information passing through thalamus

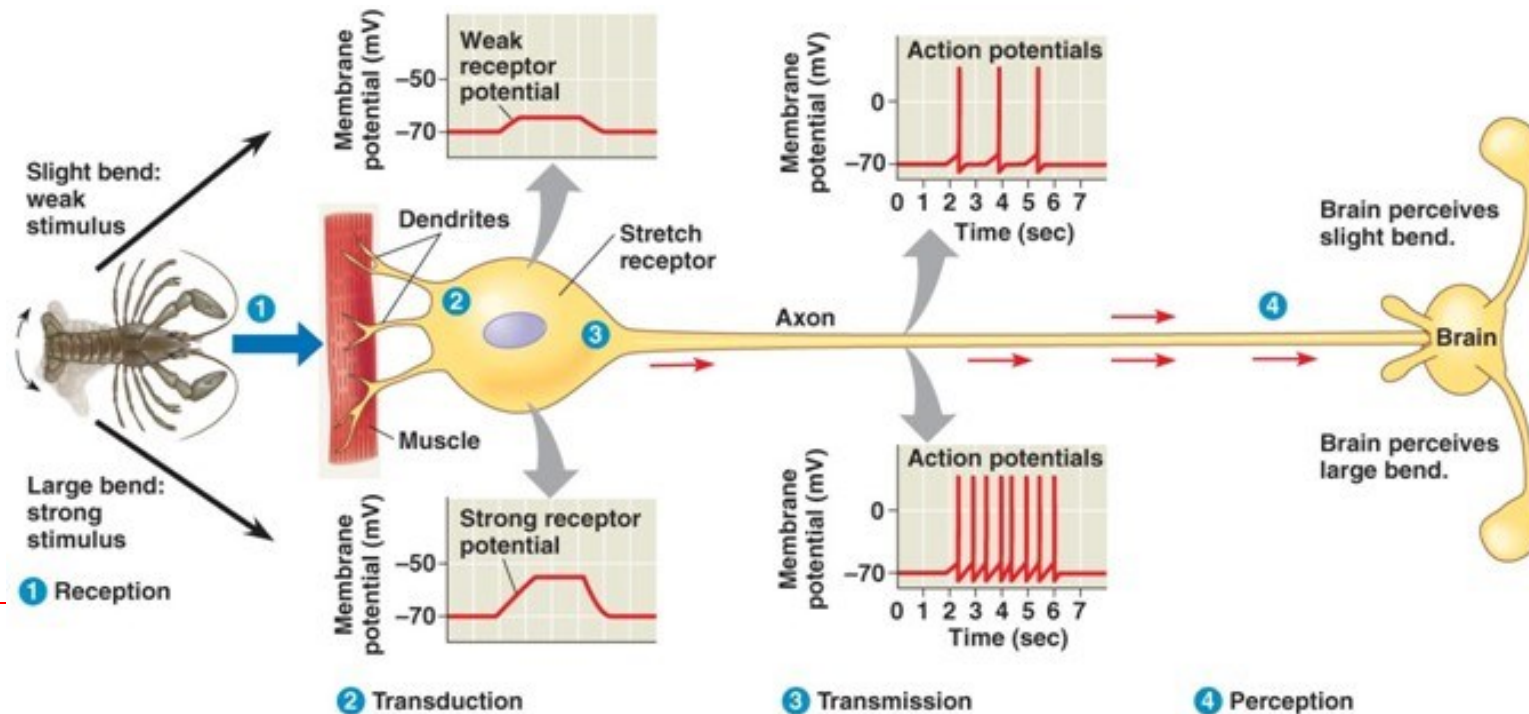
SOMATOSENSORY SYSTEM

Receptor Physiology
General/Somatic Senses
Somatosensory Pathways



RECEPTOR PHYSIOLOGY

- Transduction of sensory stimuli into nerve impulses:
- Local currents at nerve endings – **receptor potential**
 - *This is called **stimulus transduction***
- The stimulus to which a receptor specializes is called **adequate stimulus**
- Each type of receptor is highly sensitive to one type of stimulus
 - BUT any *stimuli can excite a given receptor* **if sufficiently strong**



Attributes of the stimulus

Modality (quality) discrimination = Muller's Law

- Attributes of a stimulus are mediated by *sensory system/labelled line and receptor specificity*
 - *‘labeled line principle’ or ‘law of specific nerve energies’*
- Locality sensation (Modality discrimination) depends on
 - (1) specificity of pathway labeled from the receptor till sensory cortex
 - (2) area of interpretation

No matter where along the nerve pathway one stimulates, the type of sensation will depend on which part of the brain is finally going to be stimulated.

Law of Projection:

- No matter where along the nerve pathway one stimulates, the sensation will be felt at the site of the receptor. E.g. Phantom limb sensation

	ADEQUATE STIMULUS	EXAMPLES
Mechanoreceptor	mechanical distortion	touch, pressure
Chemoreceptor	chemical concentration	taste, smell, oxygen receptor
Thermoreceptor	temperature	warm, cold
Photoreceptor	photons	visible light
Nociceptor	noxious	painful stimuli
<i>Proprioceptor (mechano-)</i>	<i>body position, muscle tone</i>	<i>joint position, tendon force</i>

Intensity discrimination

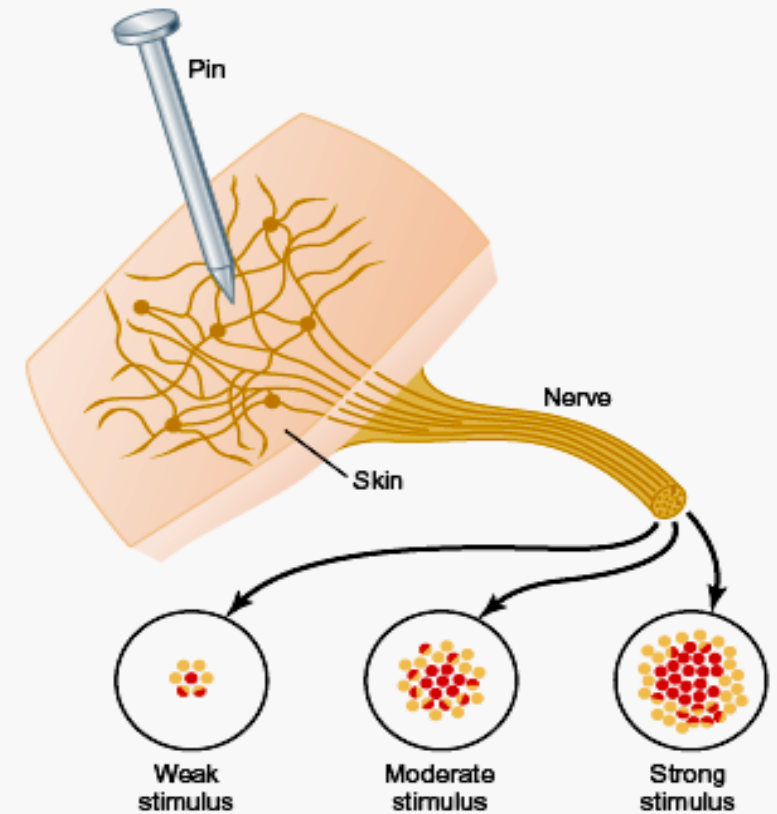
a. Weber-Fechner Law

- Intensity discrimination depends on
 - (1) *the rate of nerve impulse discharge and*
 - (2) *the number of receptors stimulated.*
- $I = K \log (S/S_0)$

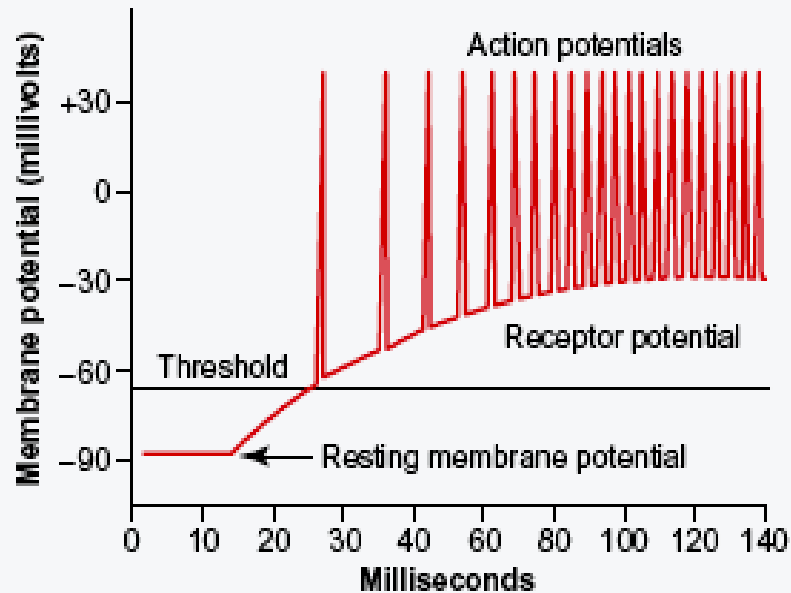
b. Spatial summation

- The greater stimulus strength, the greater is the number of fibers stimulated.

c. Adaptation

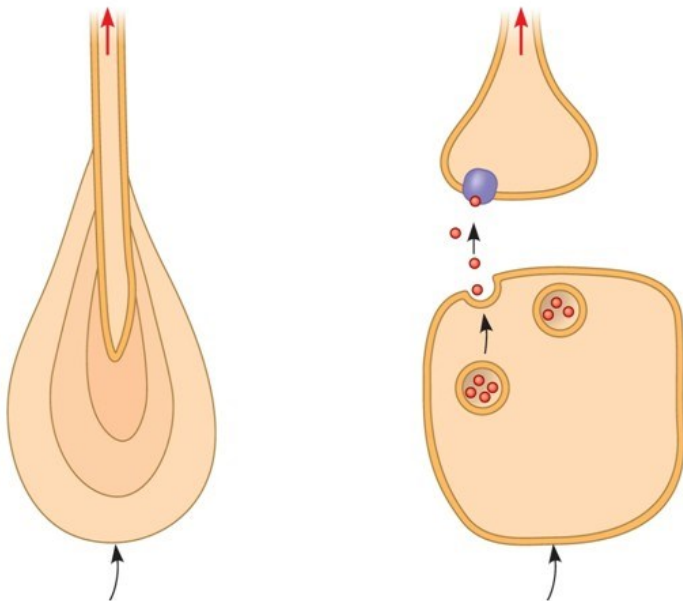


Receptor potential & AP



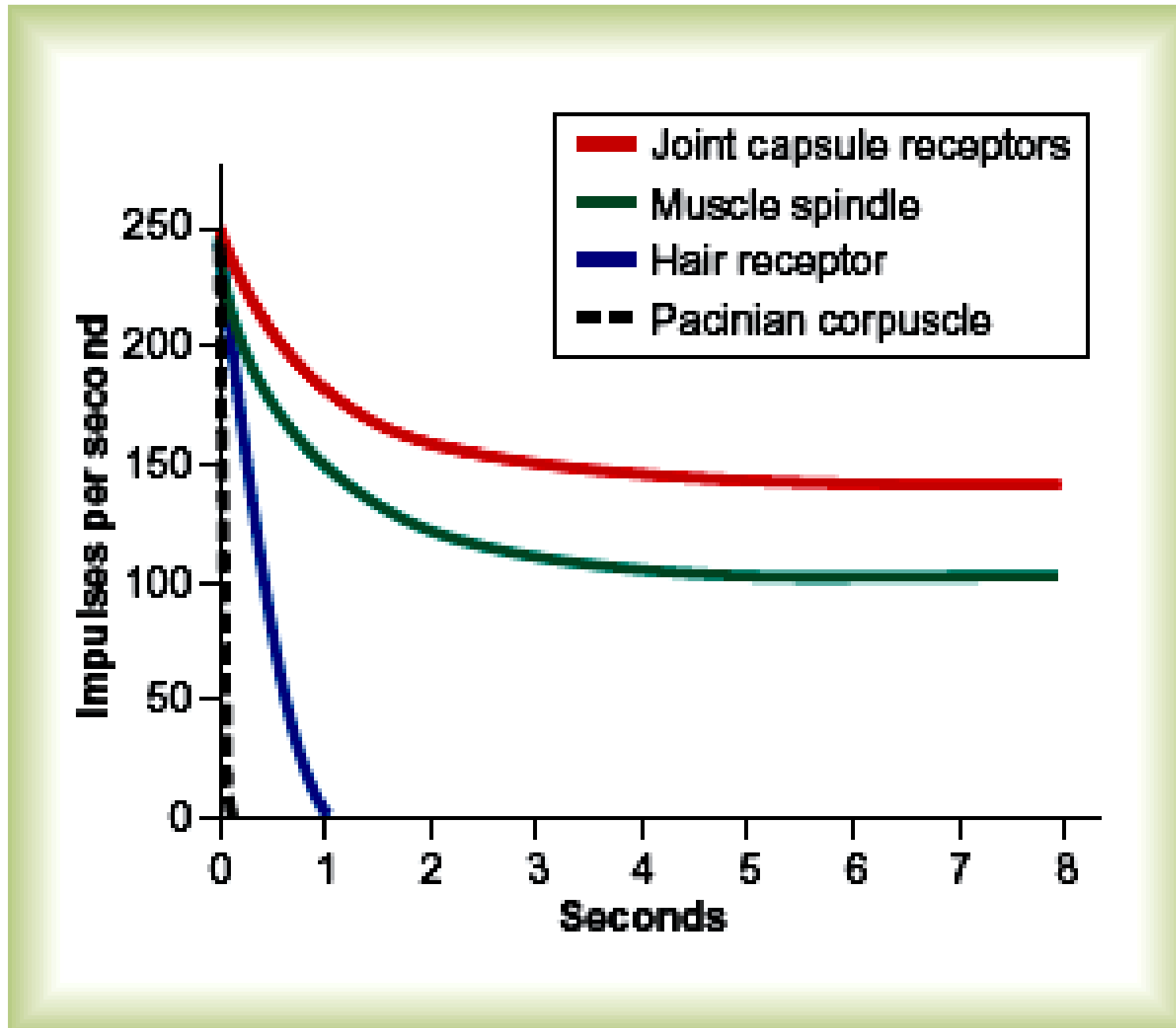
- Receptor potential is graded.
- When amplitude exceeds threshold level action potential triggers in afferent neurons (summation)
- The number of action potential is proportional to the log of stimulus intensity

- *Stronger stimuli produce more AP*



Receptor Adaptation

- Decline in number or frequency of AP to constant stimulus
- When stimulus is sustained **receptors adapt**



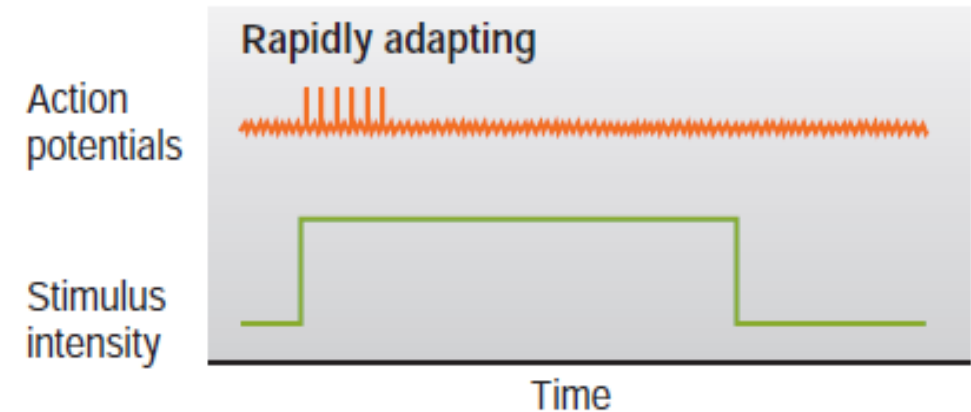
Classification of receptors based on their rate of adaptation

1. Rapidly adapting

detect rate of change in stimulus strength

(Rate/phasic/ motion/vibration receptors)

Encapsulated(Pacinian, Meissner's, Hair end organ)



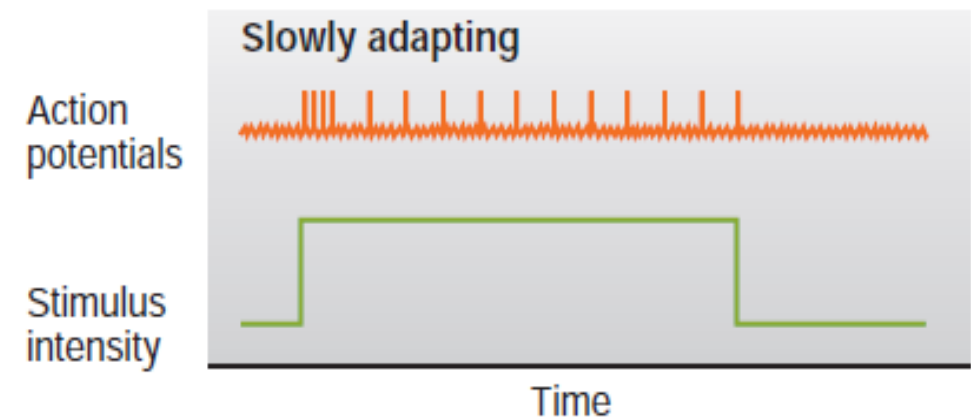
2. Moderately adapting

Ex. Thermoreceptors (20 – 40 oc)

3. Slowly adapting

detect continuous stimulus/ don't adapt to extinction *(expanded/tonic receptors)*

Ex. Proprioceptors: Muscle spindle, Free nerve ending, Ruffini, Merkel's receptors...



4. Non - adapting receptors

- keep the brain constantly appraised of the status of the body

Eg. baroreceptors & chemoreceptors of the carotid and aortic bodies; receptors of the macula in the vestibular apparatus, and pain receptors,

Location

- **Basis:** location of the sensory receptor and anatomical (topographic) organization of sensory pathways
- **Law of Projection:** sensation is felt as arising from (projected to) the receptor's receptive field even when it arises elsewhere

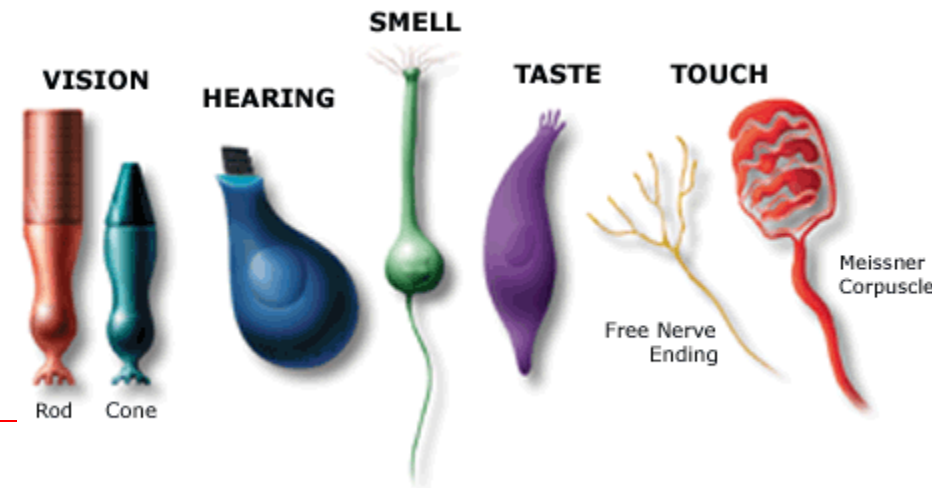
Ex. phantom limb

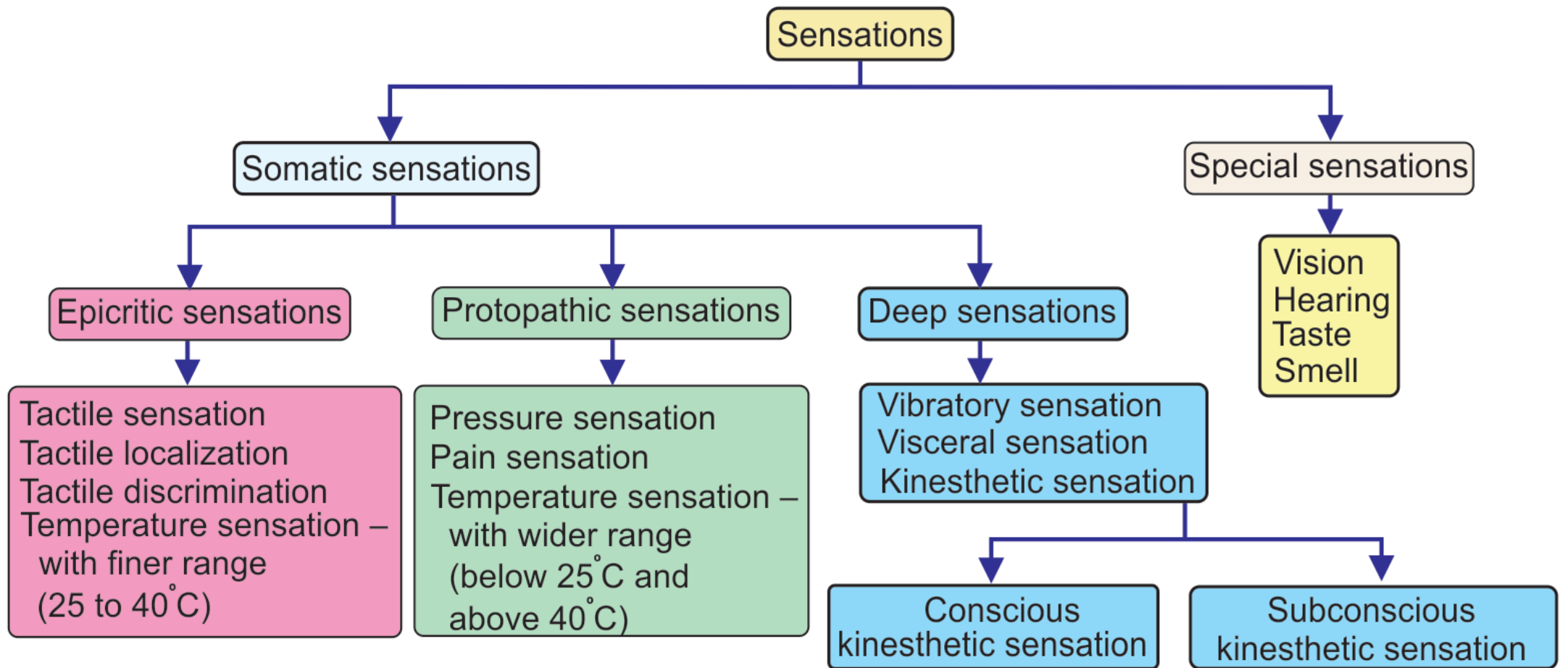
- **Acuity:** the precision of stimulus localization or the ability to distinguish fine details depends upon
 - *size of the receptive field (small field $\Rightarrow$ better acuity)*
 - *innervation density (higher density $\Rightarrow$ better acuity)*
 - *convergence along CNS pathways (less convergence $\Rightarrow$ better acuity)*
 - *lateral or surround inhibition (increases acuity; discussed later)*

○ *Tactile acuity is slightly greater in women than in men and varies between fingers but not between hands. The distal pad of the index finger has the max sensitivity.*

Somatic sensation

- Sensations: *Somatic, visceral, special...*
 1. *General Visceral Afferent (GVA)* – from viscera
 2. *General Somatic Afferents (GSA)* include sensation from
 - *skin (somesthetic sensation)* and
 - *deep structure (proprioception sensation)*
- Relatively simple endings (compared with special senses)
- Brain decode the type of modality based on *the labeled line principle*
 - Because each modality has specific
 - **Receptor (adequate stimulus)**
 - **Pathway**
 - **Area in the cortex**





Classification of somatic sensations

General classification

a. “Discriminative”

*encapsulated endings innervated
by larger myelinated axons;
greater sensitivity;
more precise localization*

b. “Crude”:

*free nerve endings
less sensitive; less precise
localization*

Nature/Major modalities

1. Mechanoreceptive

- Tactile
 - Touch
 - » Crude, Fine
 - Pressure
 - » Deep, Muscle tension
 - Vibration
- Kinesthetic
 - Sense of position
 - Sense of movement

2. Thermoreceptive

3. Nociceptive (plymodal/pain/itch)

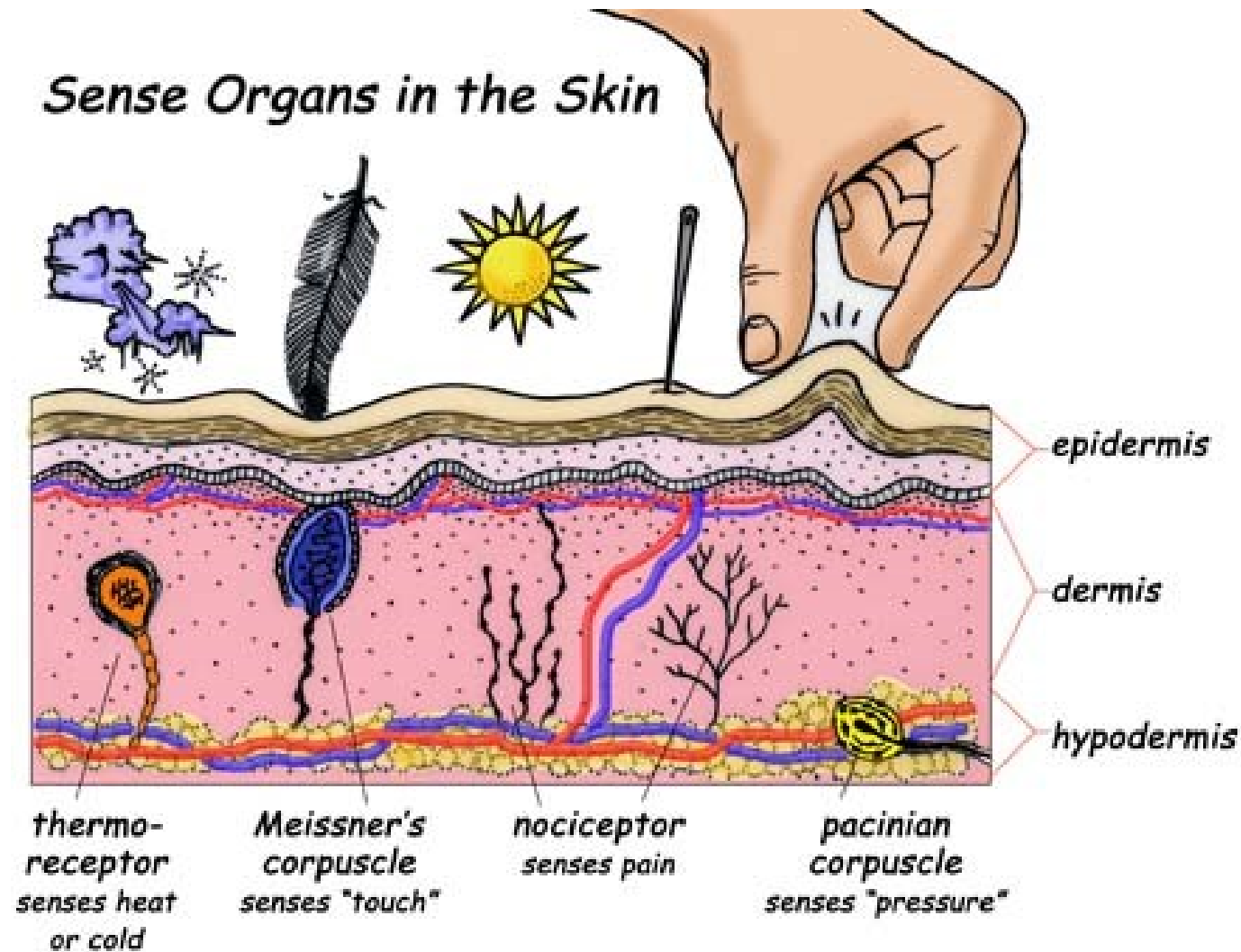
According to their location

1. Superficial (cutaneous)

- Pain
- Touch
- Temperature

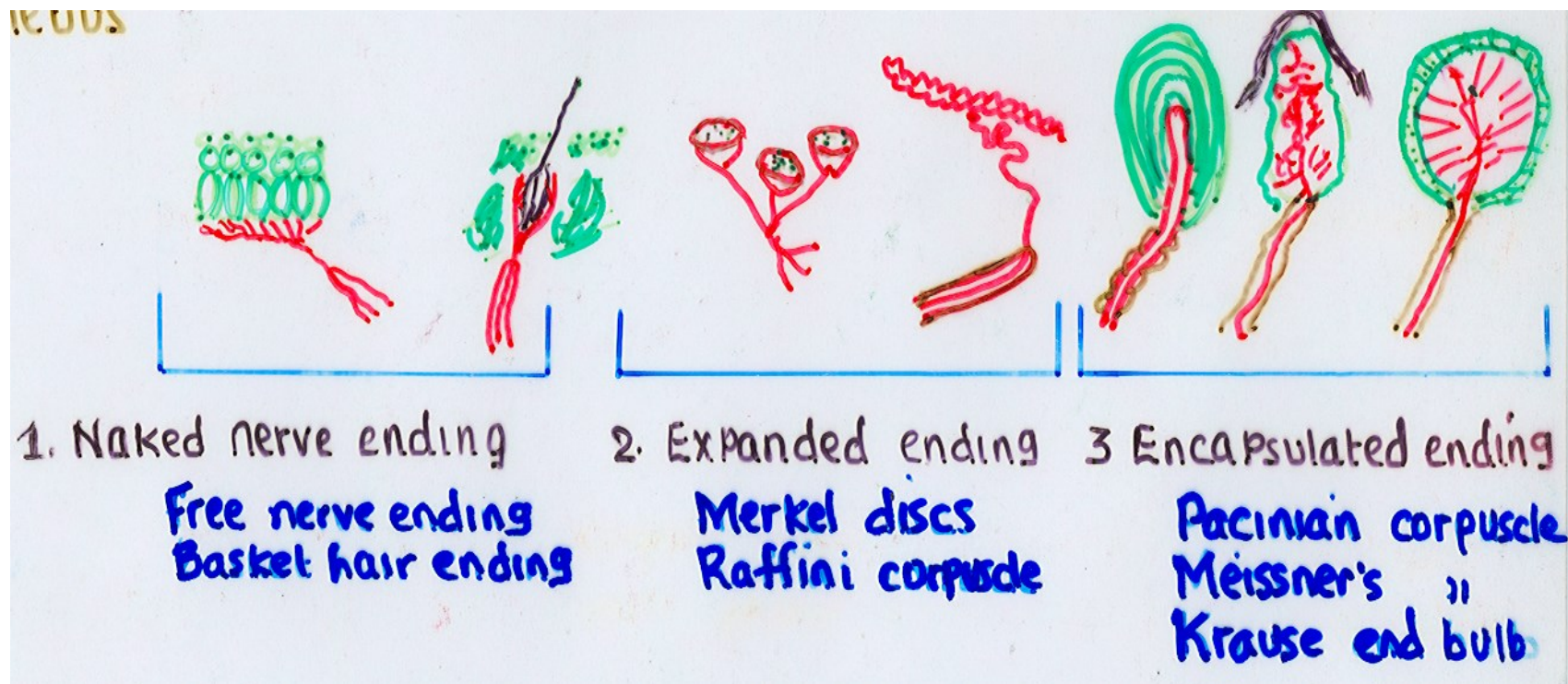
2. Deep (proprioceptive)

- Kinesthetic
- Pressure
- Muscle tension/GTO/
- Muscle stretch /spindle



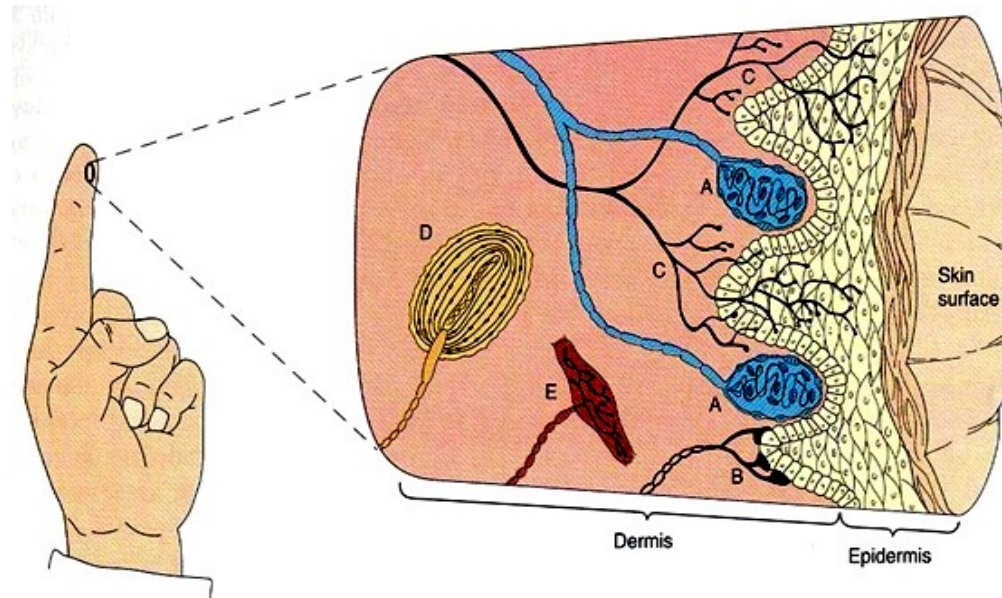
Cutaneous receptors

A. Mechanoreceptors: Cutaneous



	<u>Fibers</u>	<u>Adequate Stimulus</u>	<u>Field Size</u>	<u>Adaptation</u>	<u>Sensation</u>
Merkle c " Edge/points	A β	Skin indentation	small	slow	Touch-pressure
Meissner corpuscle	A β	Vibration	small	rapid	Flutter, contact
Ruffini ending	A β	Skin indentation	large	slow	Touch stretch
Pacinian corpuscle	A β	Vibration	large	very rapid	Vibration
Hair follicle	A β	Hair bending	small	rapid	Touch, contact
Free ending	A δ	Skin indentation	large	rapid	Contact (coarse) (Itch/tickle)
Muscle spindle	Ia, II	Muscle length		rapid, slow	Proprioception
Golgi tendon organ	Ib	Muscle tension		slow	Proprioception
Joint receptors	small	Joint angle		rapid	Proprioception

Mechanoreceptors in Glabrous (non hairy) Skin



- A - Tactile (Meissner's) corpuscle
- B - Tactile (Merkle's) corpuscles
- C - Free terminal
- D - Lamellated (Pacinian) corpuscle
- E - Ruffini corpuscle

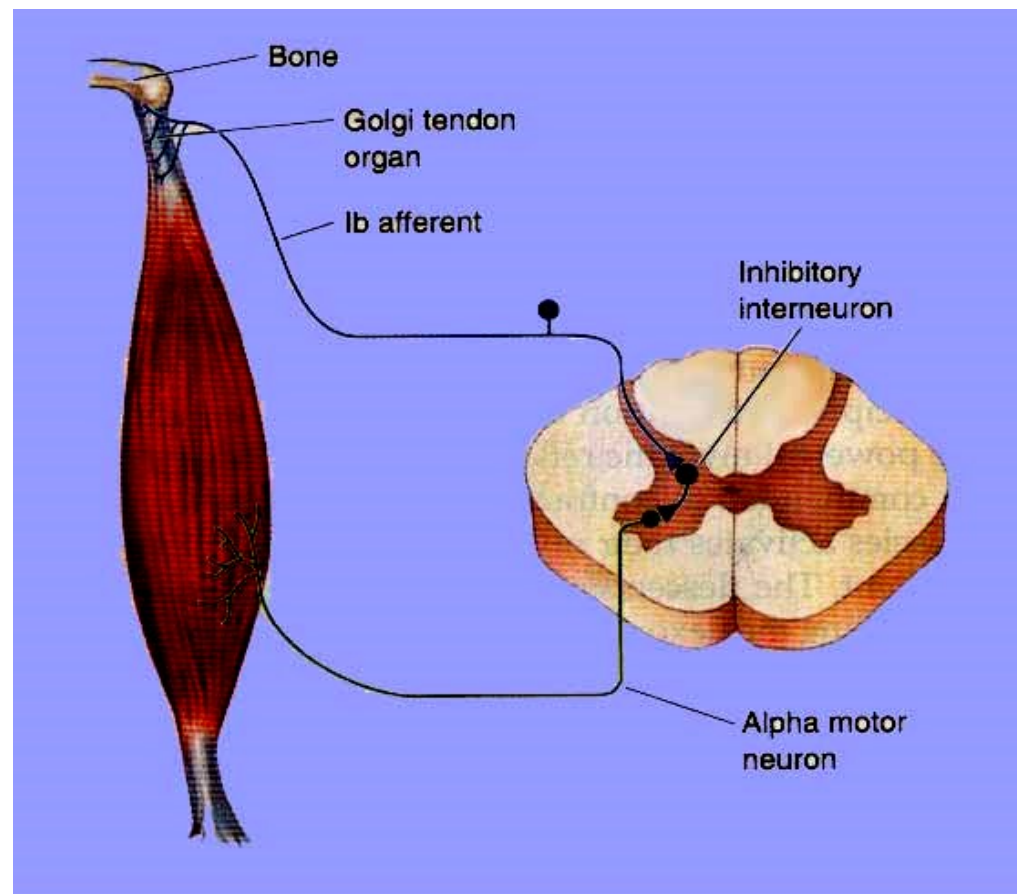
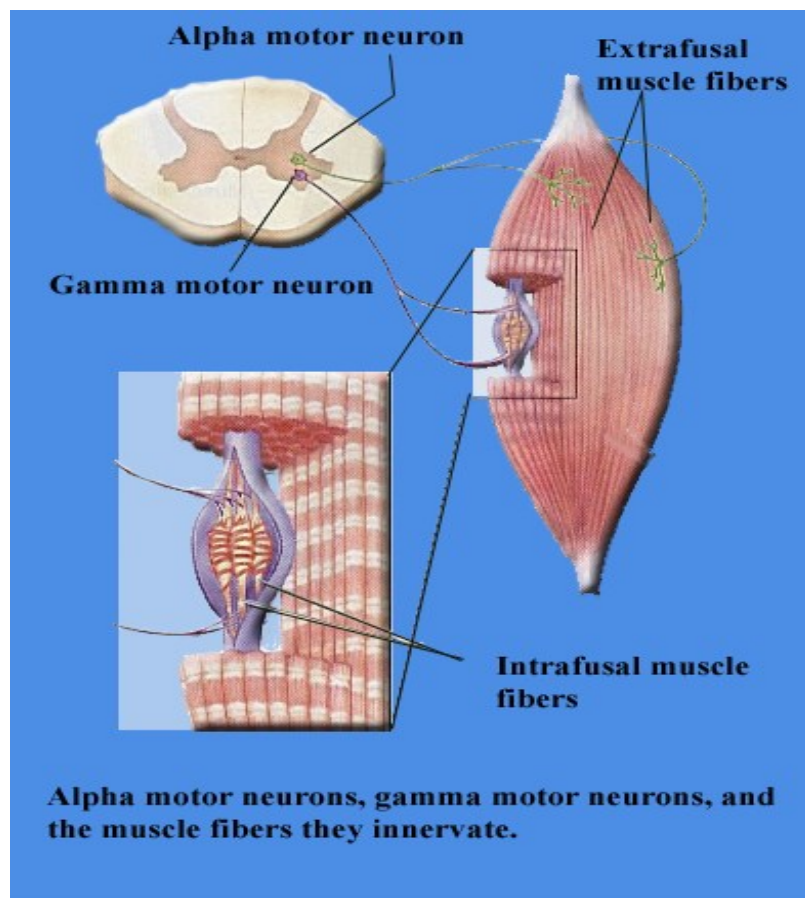
Rapid
adaptation

Slow
adaptation

Superficial	Deep
Small field	Large field
M eissner's Corpuscle	Pacinian Corpuscle
M erkel's Disc	Ruffini End Organ

2. Deep mechanoreceptors/Interoceptors (*DCML, pathways to cerebellum*)

1. **Muscle spindles**: stretch receptors in the fleshy part of the muscle
 - *Nuclear bag fibers/annulospiral/primary endings*: onset muscle stretch
 - *Nuclear chain fibers/flower spray/secondary endings*: Stretch on progress
2. **Golgi tendon organs**: tension receptors in the tendons of skeletal muscles
3. **Joint-capsule receptors**
 - Located in and around joints.
 - Gathers information about limb position and joint movement.



Mechanoreceptive sensations

Tactile sensation

I. Crude touch

- Locality identification is poor
- Receptors are *free nerve endings* and *basket hair endings*
- Use *A-delta fibers (A δ fibers)* = ($d = 1-5 \mu m$, $v = 5-15 m/s$).
 - *Tract:* spinal cord $\rightarrow$ ventral spinothalamic tract $\rightarrow$ somatosensory cortex.

II. Fine touch sensations

- Receptors: *Meissner corpuscles, and Merkel's disc*
- Nerves: *A-beta nerves*
- Accurate in the fingertips and lips than in other areas.
- (Rich in receptors + widely represented in S1)

1. Tactile localization (topognosis)

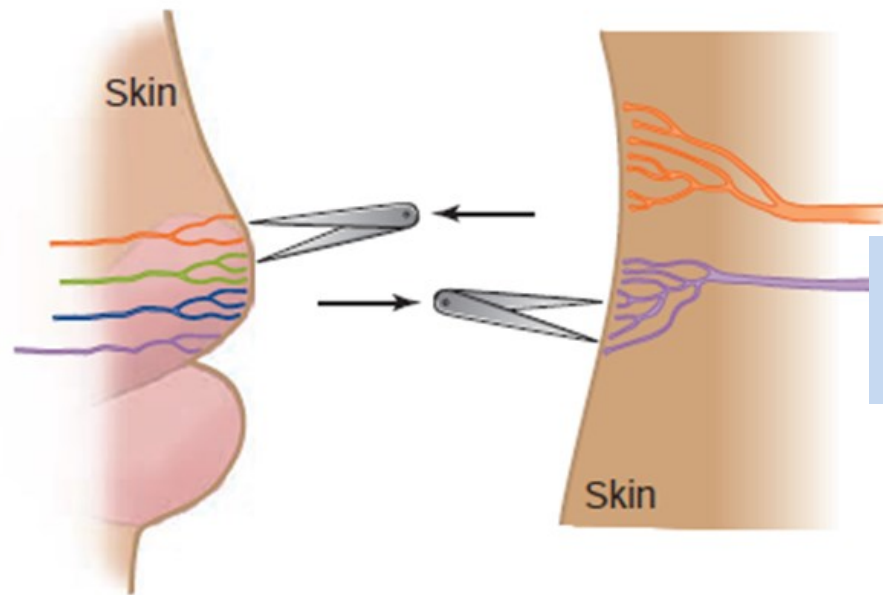
- Ability to identify where stimulus is applied

2. Tactile (two-point) discrimination:

- The ability to identify two tactile stimuli applied simultaneously as two separate point of contact.

Lips: Two distinct points are felt

Back: Only one point is felt



1-2 mm in finger tips, 30-70 mm in back

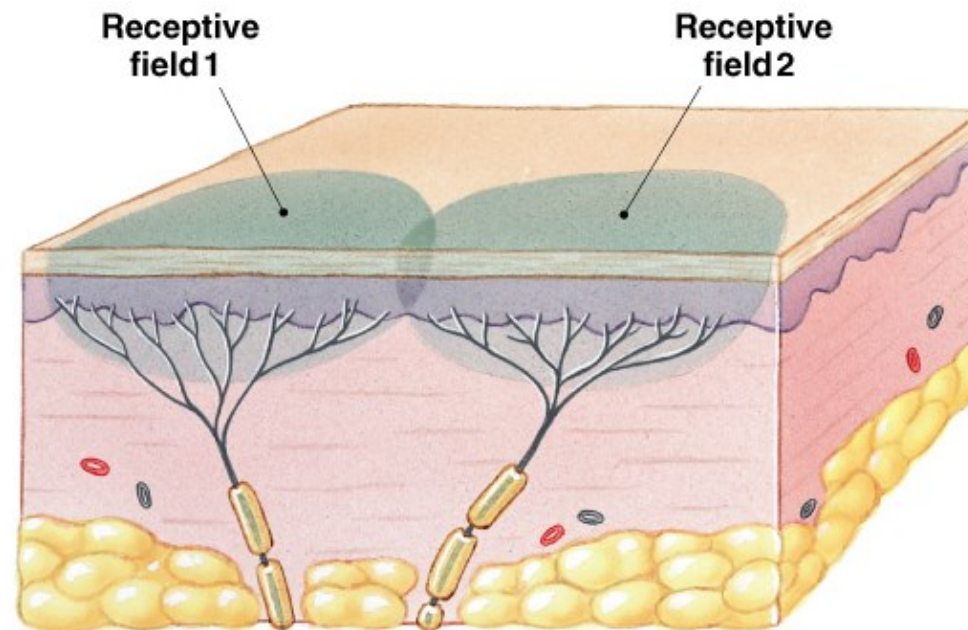
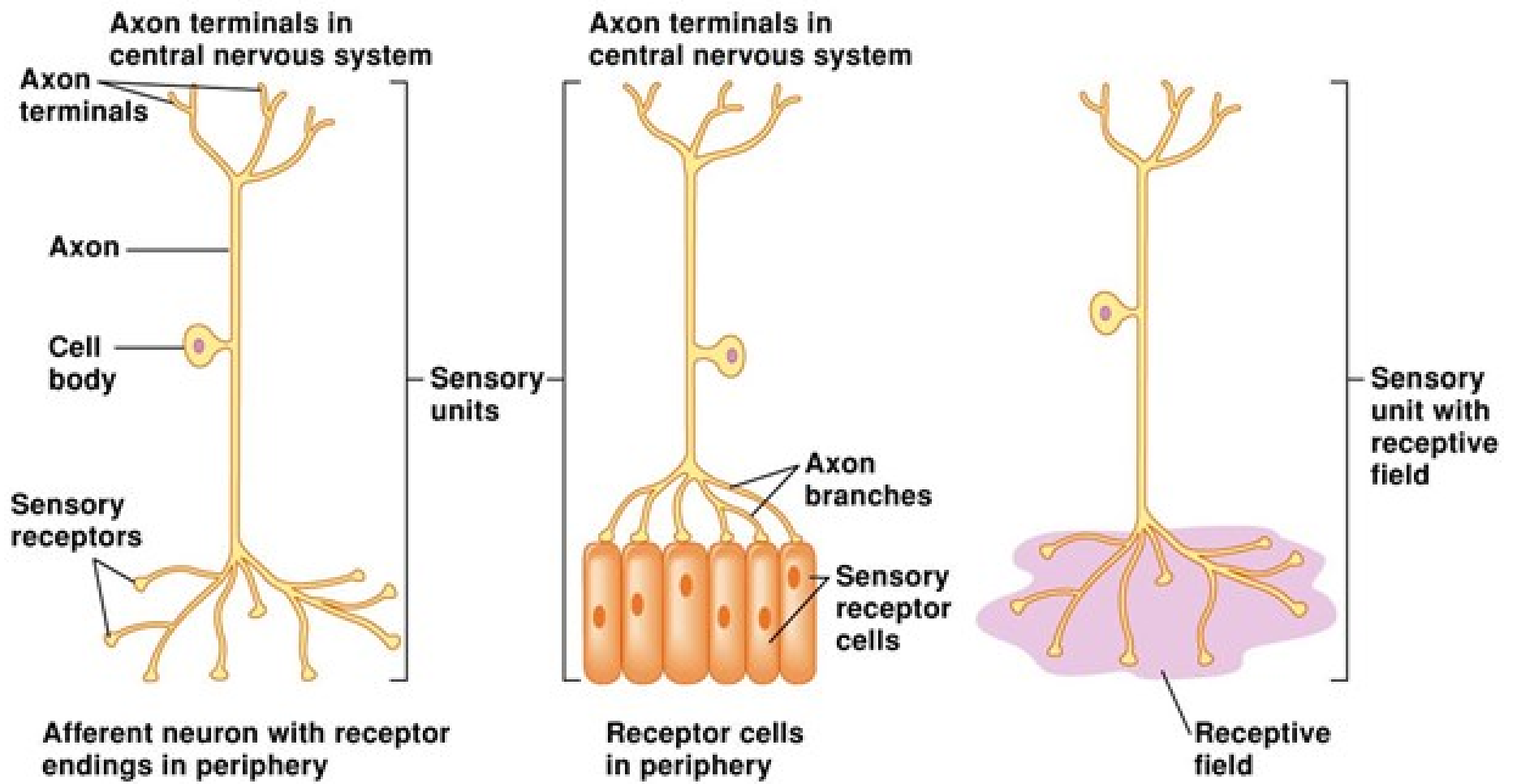
3. Stereognosis:

- The ability to recognize the nature of objects by handling them (from their shape, size, weights....).
 - Relies heavily on the integrity of tactile/pressure sensation and also cortical sensory centers.*

SENSORY SYSTEM OF THE BRAIN

- **Tract**
 - Nerve fibers having common origin, pathway, destination and function
 - *Ascending tract*- sensory function
 - *Descending tract*- motor function
 - Sensory tract originate in the dermatome as processes of *the first order neuron*.
 - *Receptive field*: Sensory endings innervated by afferent axons in nerve
 - *Dermatome*: Narrow band of skin that receives somatic afferent of spinal nerve or axons of a single dorsal root(*Cutaneous receptive fld*).

Deep and visceral sensory axons join with cutaneous axons to form dorsal root



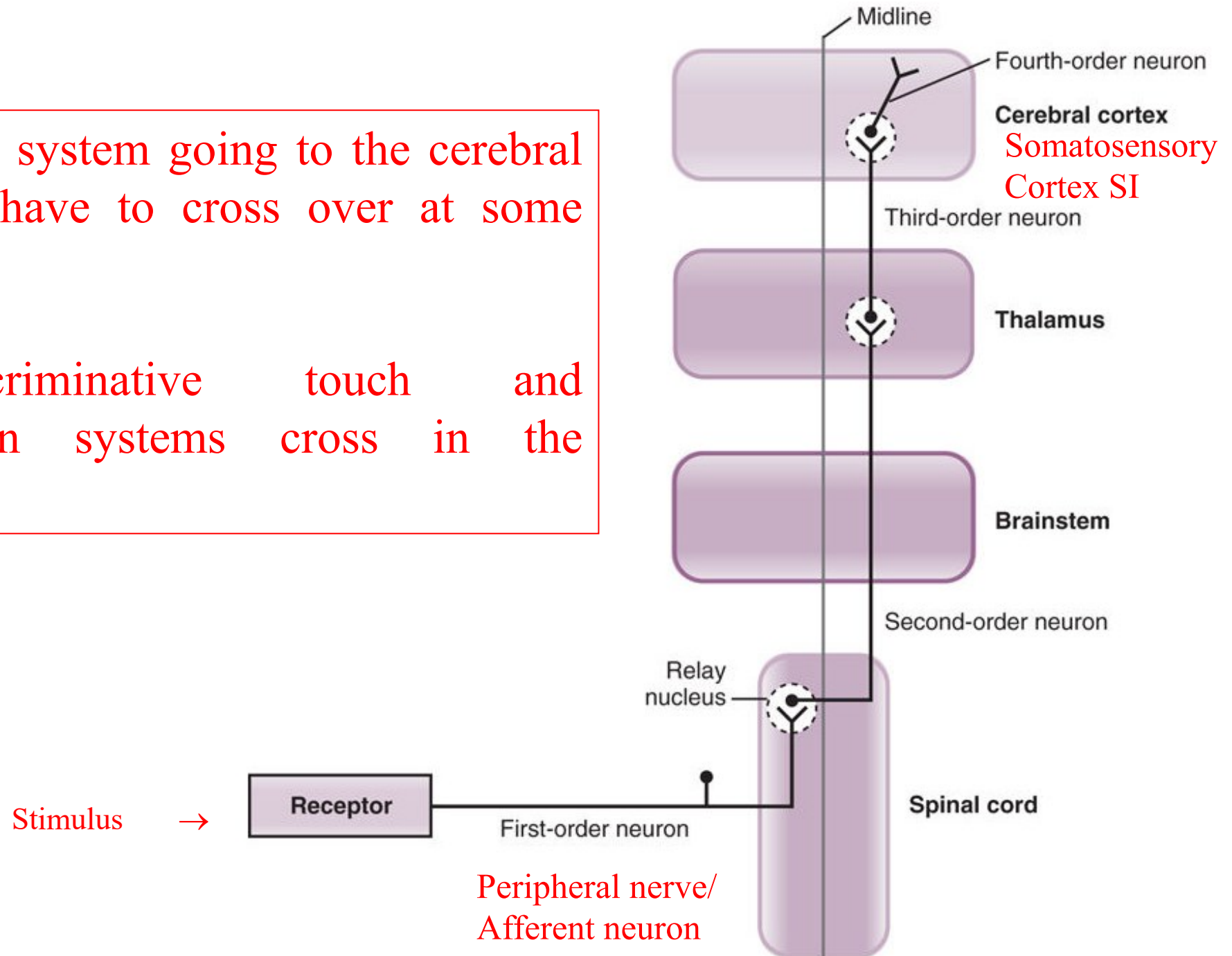
SOMATOSENSORY PATHWAYS

(Ascending tracts)

- An *ascending tract* is a sensory tract, which conducts sensory signals up to the higher brain centers.
- Starts from sensory receptors → sensory neurons in the brain.
- Involves 2-4 orders of neurons ($1^{\text{st}} \rightarrow 2^{\text{nd}} \rightarrow 3^{\text{rd}} \rightarrow 4^{\text{th}}$).
- First order neurons are the afferent sensory nerves (cell bodies in the dorsal root ganglia/cranial nerves ganglia).

•1 st order neuron:	Dorsal root ganglion (spinal/trigeminal ganglion)
•2 nd order neuron:	Spinal cord/medulla
•3 rd order neuron:	Thalamus VPLNT
•Destination:	Cerebral cortex/Postcentral gyrus (Primary sensory area I)

- Any sensory system going to the cerebral cortex will have to cross over at some point.
- The discriminative touch and proprioception systems cross in the medulla.



Pathways for somatic perception

project to the somatosensory cortex and cerebellum

- Somatosensory modalities: *Touch, Proprioception, Temperature, Nociception/Pain / Itch.*
- Based on entry point in the spinal cord/brain, ascending tracts are
 - **Dorsal column-medial lemniscus pathway**
 - **Gracile (<T6) and cuneate (>T6) pathway** (fine touch, pressure)
 - **Anterolateral pathway**
 - *Ventral/Anterior spinothalamic tract* – Crude touch/itch/tickle
 - *Lateral spinothalamic tract* – Pain and temperature
 - *Paleospinothalamic* – Slow pain and temperature
 - *Neospinothalamic* – Fast pain
 - **Spinocerebellar pathway**
(also spino-olivary, spino-vestibular tracts – proprioception)

Two Ascending Sensory Pathways

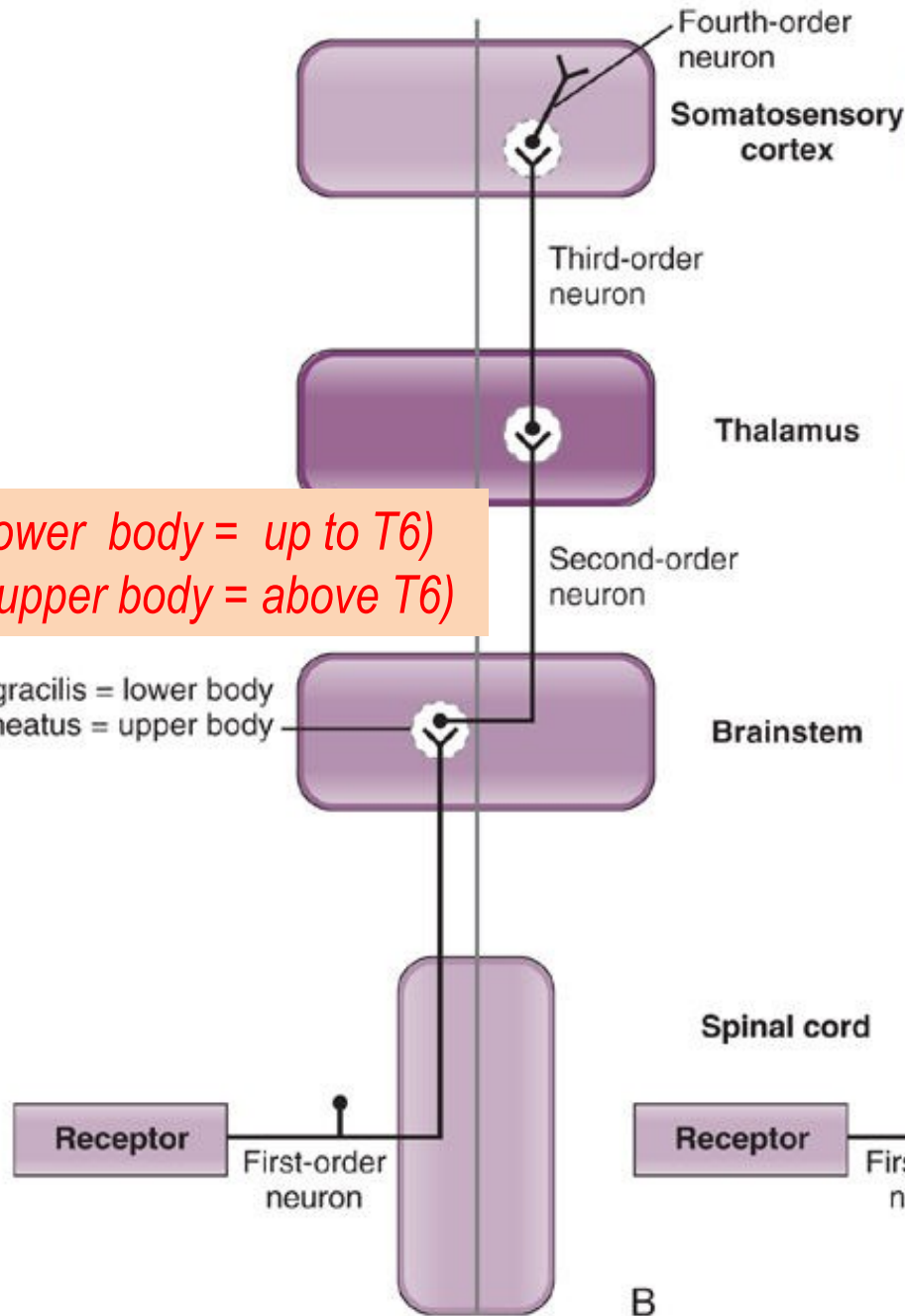
(fine touch, pressure, proprioception)

(pain, temperature, crude touch)

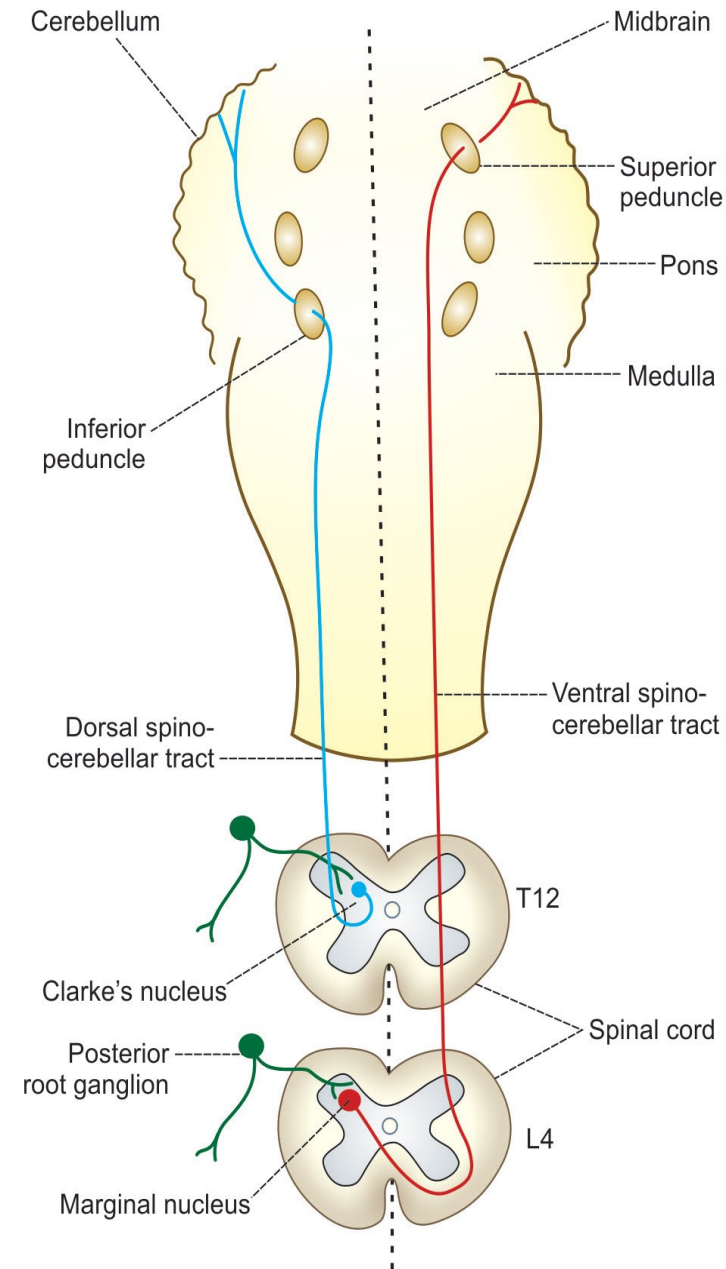
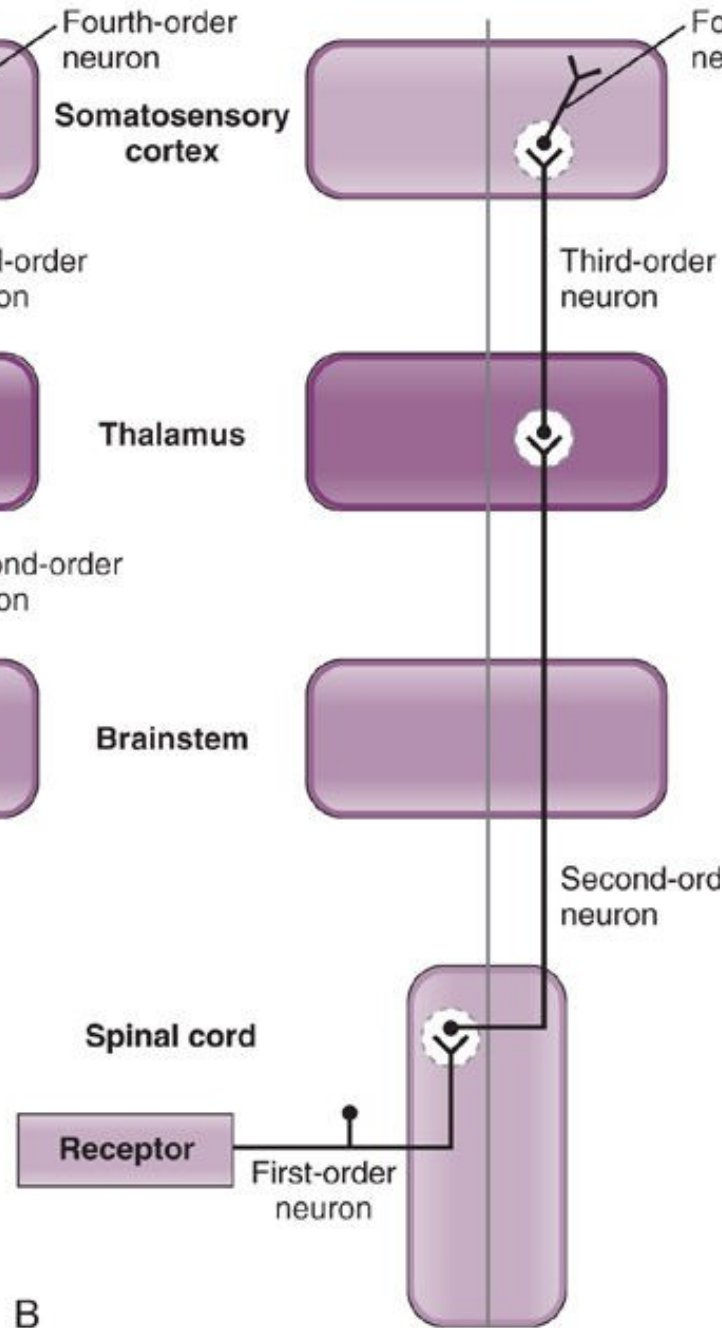
Gracile (lower body = up to T6)
Cuneate (upper body = above T6)

Nucleus gracilis = lower body
Nucleus cuneatus = upper body

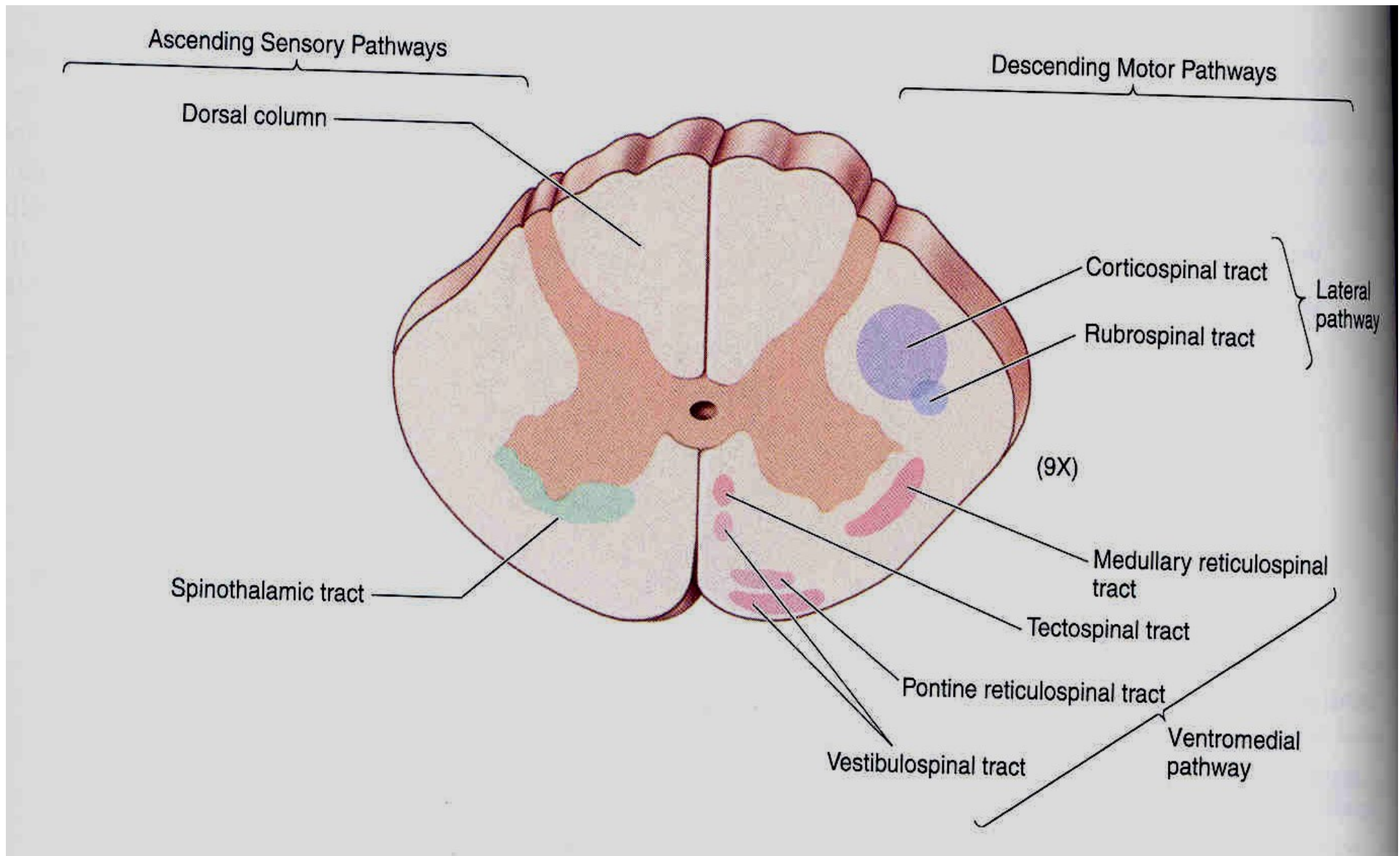
A



B



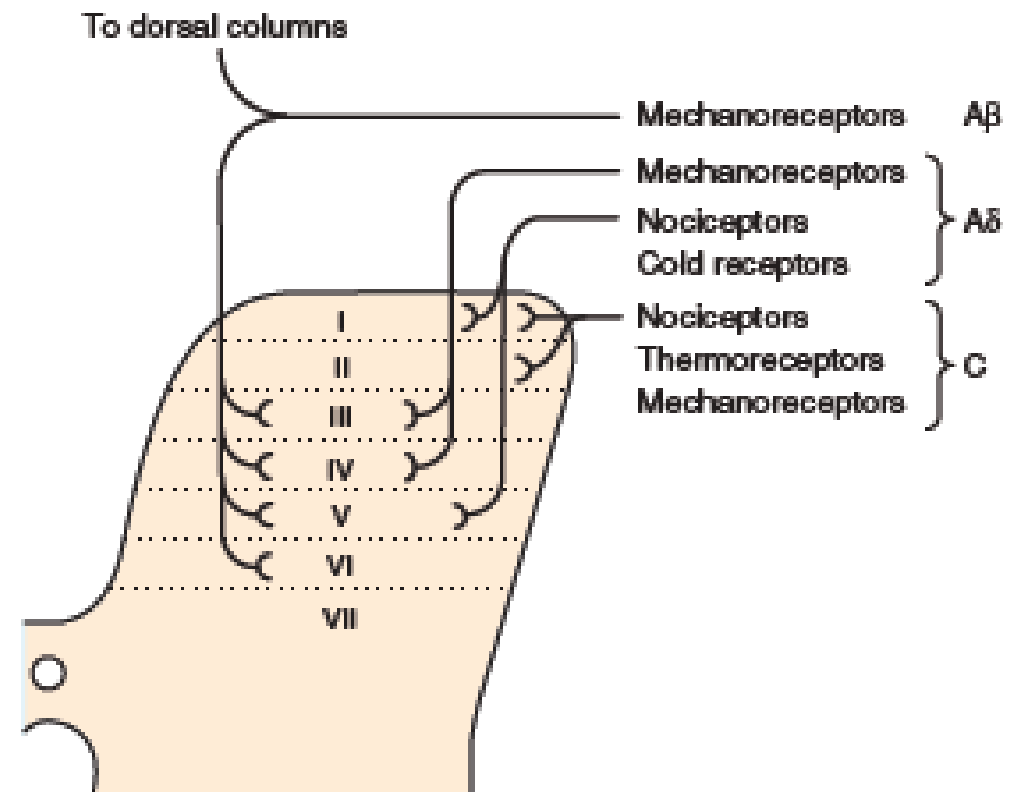
Cross sectional study of spinal cord



Spinal component of the somatosensory system

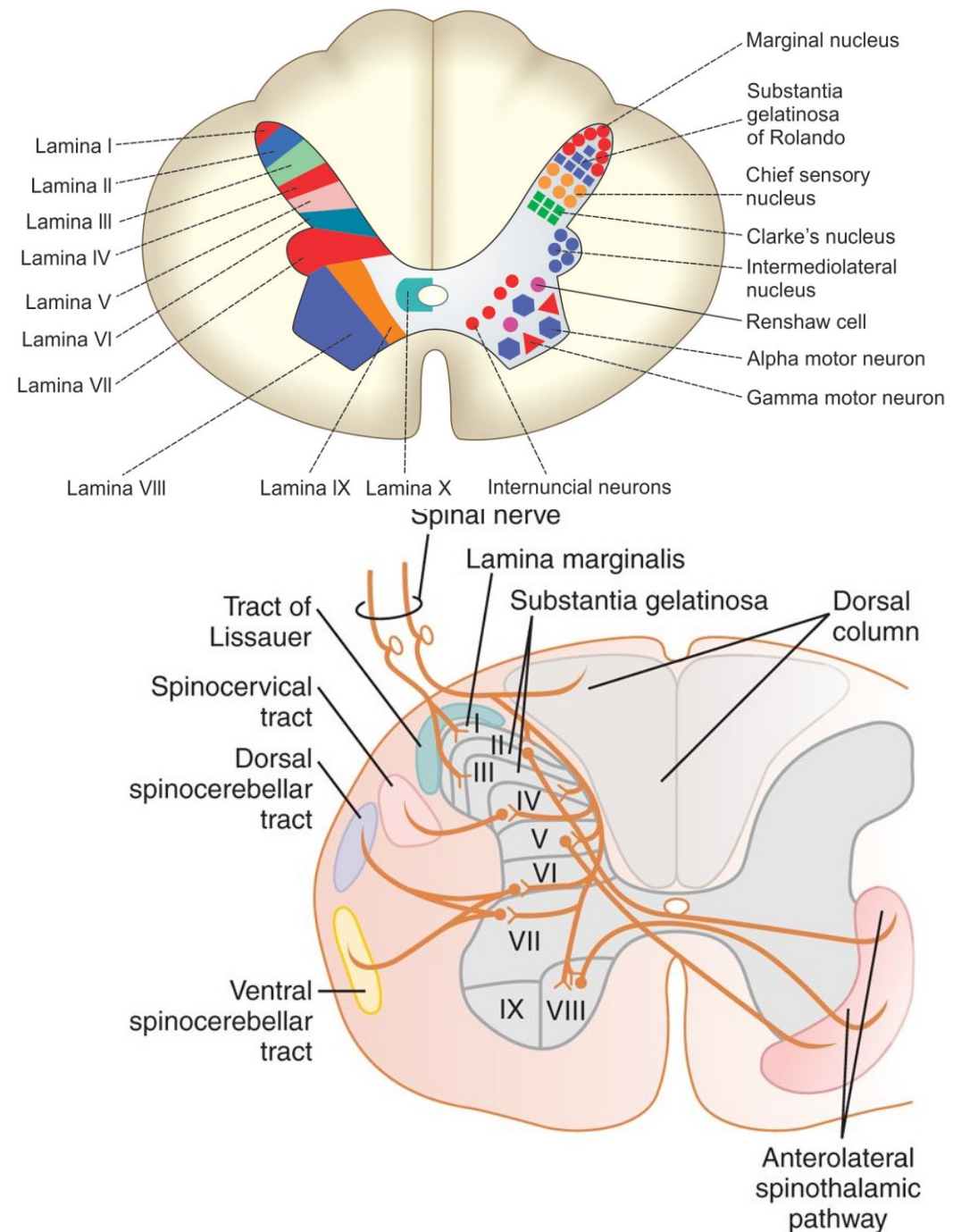
Modality segregation

- Physiological and morphological properties of axons of DRG
- Sensory information from *mechenoro-noci-thermoreceptor* terminate in a unique group of neurons in the spinal cord



The course of sensory nerves in spinal cord

- Ascend or descend few segments in the dorsal horn forming **Lissaur tract**.
- Relay in neurons of the different **spinal lamina** gray matter.
- Proceed on to **directly relay** on motor neurons of the ventral horn (reflex arc).
- **Ascend without relay** up to medulla in the dorsal column



Bell-Megendie Law: the dorsal root is sensory and ventral root is motor.

Thalamus as component of the somatosensory system

- Thalamus

- Large ovoid mass of gray matter - **each thalamus has 5 nuclei groups**
- All nerve signal that go to cerebral cortex pass via thalamus (*secretary of cerebral cortex*) *except olfaction*
- Connects brain areas with each other (integrating, modulating + gating, bidirectional flow of information).
- Motor integration (inputs from cortex, cerebellum + basal ganglia).
- Pain modulation (all nociceptive information).
- Arousal (part of **ARAS**).
- Part of the limbic circuit (recent/declarative memory).

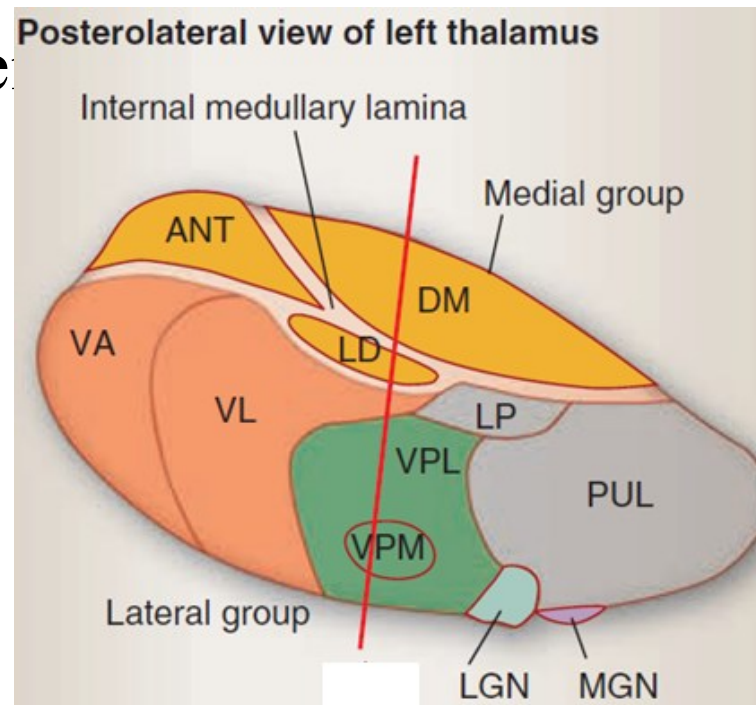
Anatomical division of the thalamus

1. Anterior group

1. Connected to *cingulate gyrus*
2. Form part of the *limbic lobe*
3. Concerned with recent memory

2. Medial group

1. Dorsomedial nuclei
2. Intralaminar nuclei
3. Middle nuclei



3. Lateral nuclei

Dorsolateral
Posterolateral

4. Ventral group

Ventroanterior
Ventrolateral
Ventroposterior
(ventrobasal)
Ventroposterolateral
Ventroposteromedial

5. Posterior group

Medial geniculate body
Lateral geniculate body

Functional division of the thalamus

- 1. Specific projection nuclei (cortical relay)
 1. VPL - somatosensory pathway from *trunk and limbs*
 2. VPM - trigeminal sensory pathway from *head*
 3. *MGB - relay of auditory pathway*
 4. *LGB - relay of visual pathways*
- 2. Non specific nuclei
 - 1. Intralaminar nuclei
 - 2. Ventroanterior nuclei
 - 3. Anterior nuclei
 - *Afferent – receive from ascending reticular activating system*
 - *Efferent - to hypothalamus, cerebral cortex and limbic system*
 - *consolidation of memories, motivation, and direction of attention to specific stimulus.*

3. Association nuclei

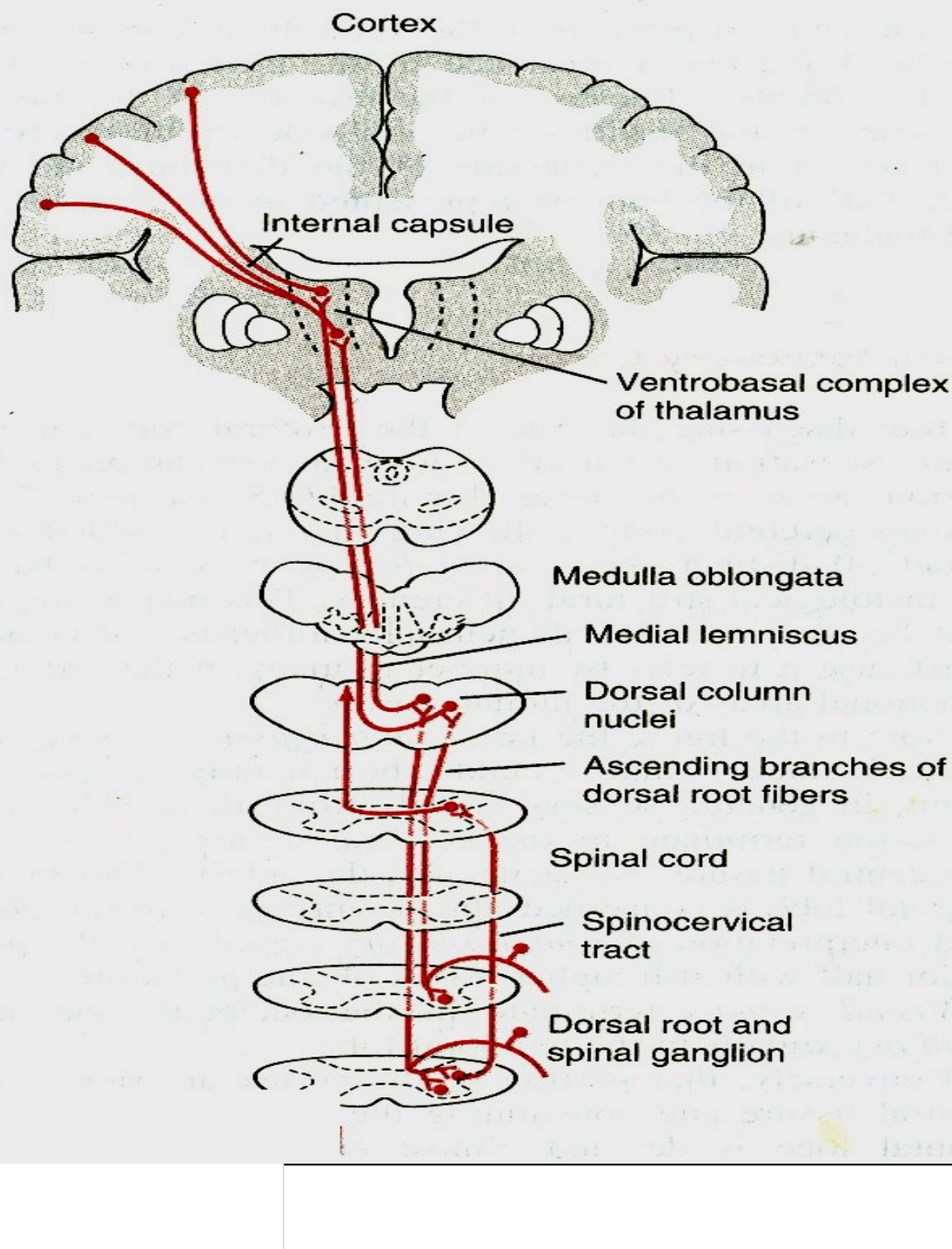
- Dorsomedial (DM), Dorsolateral, Posterolateral
 - *Afferent- from thalamic nuclei*
(BG, brainstem, limbic system)
 - *Efferent- Only to cortical association area*

5. Motor nuclei

- Ventrolateral nuclei
 - *Afferent – From basal ganglia & cerebellum*
 - *Efferent - To motor cortex*

THE DORSAL COLUMN PATHWAY (MEDIAL LEMNISCAL)

Gracile (*lower body = up to T6*) and **cuneate** (*upper body = above T6*) tract



- $A\alpha$, $A\beta$ thick myelinated fibers (Rapidly conducting fibers).
- Conducts sensations from the ipsilateral side of the body (till medulla).

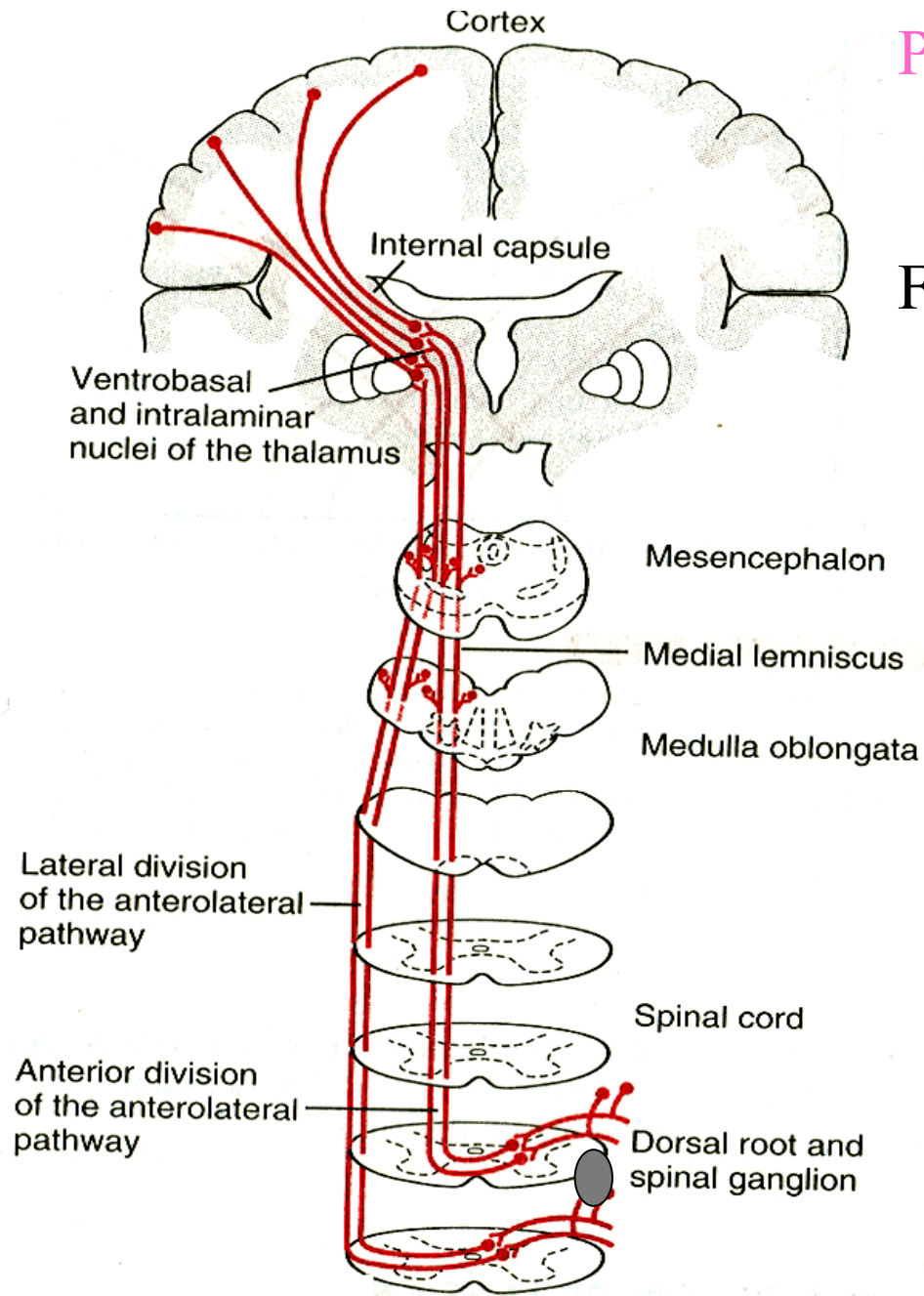
Fine touch

- Fine pressure
- Vibration (Pacinian c.)
- Stereognosis (touch+pressure)
- Muscle tension
- Proprioception (kinesthesia)

↑ Tactile localization & discrimination

THE ANTEROLATERAL SYSTEM

1. Ventral (anterior) spinothalamic pathway

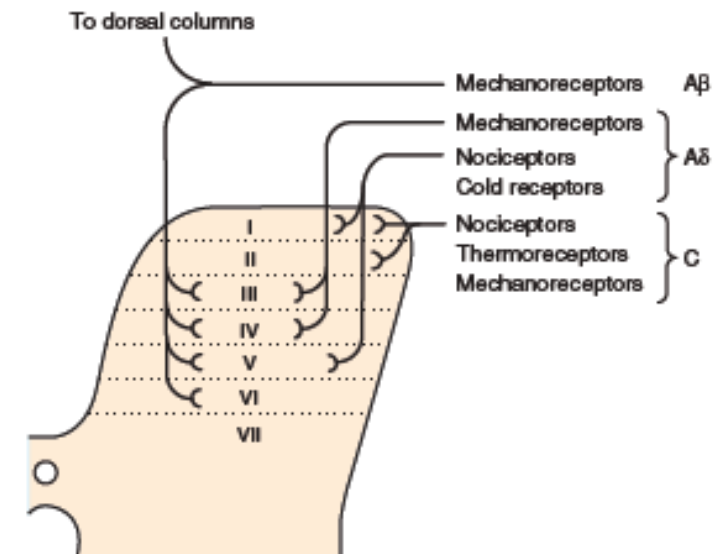


Pathway for prothopatic sensation

- Crude touch, crude pressure & itch

Four orders of neurons

- *Neurons of DRG*
- *Lamina II, III, IV*
- *VPLN of thalamus*
- *Somatosensory cortex*



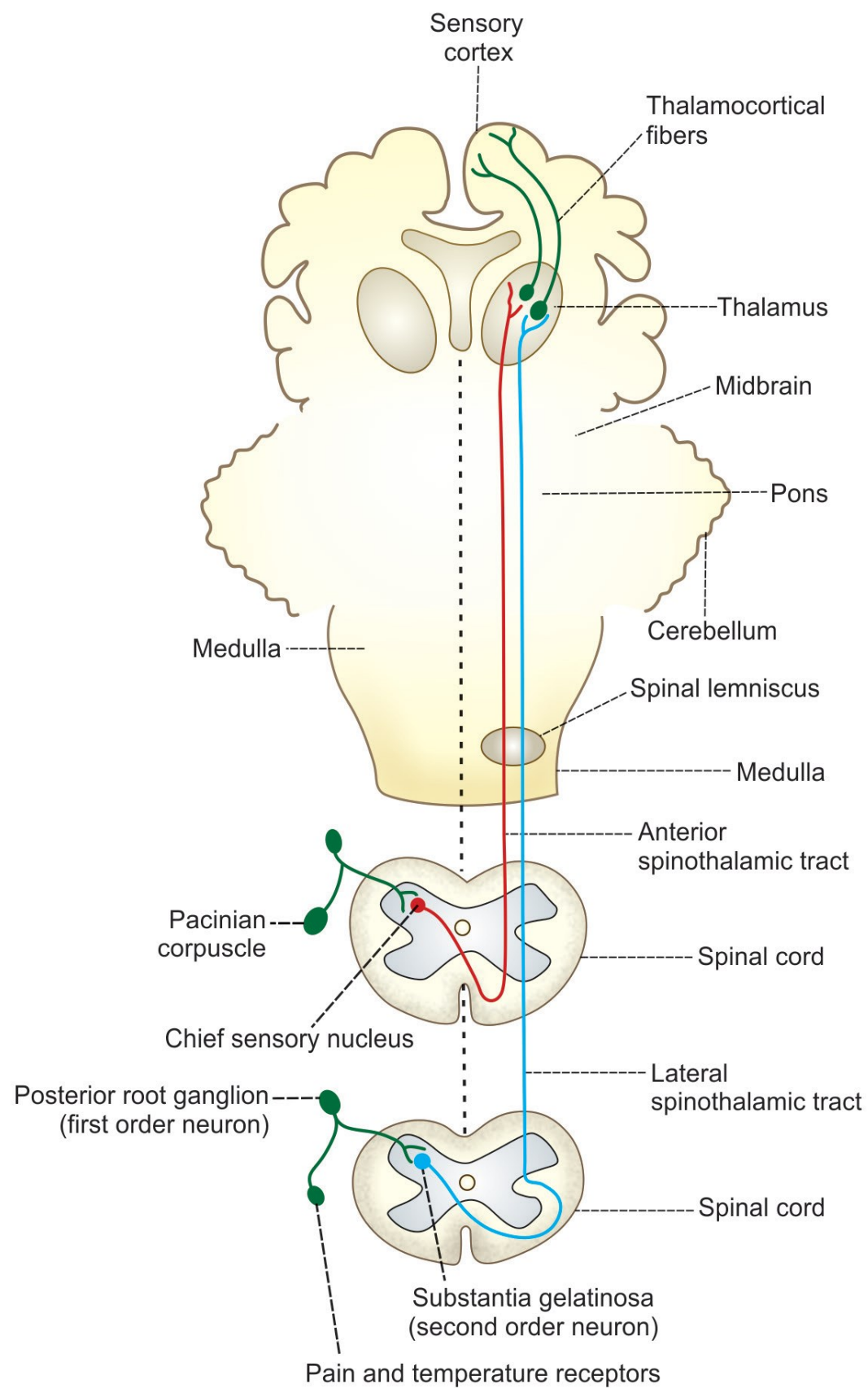
2. The lateral spinothalamic pathway

- Temperature sensation
- Pain sensation: *Slow (pathological) and Fast (physiological)*

SLOW PAIN + T°

Paleospinothalamic pathway

- Neurons of DRG / C fibers terminate on substantia gelatinosa/
II, III
 - **Intralaminar nuclei** (Thalamus, non specific)
 - All sensory neurons of cerebral cortex
(reticular formation, pons, limbic system, and mid brain)
 - Influence emotional & visceral responses to pain + descending modulation of pain.
 - Nociceptors are *non-adaptive* thus suppression is done at central level
- *Endogenous analgesia system*

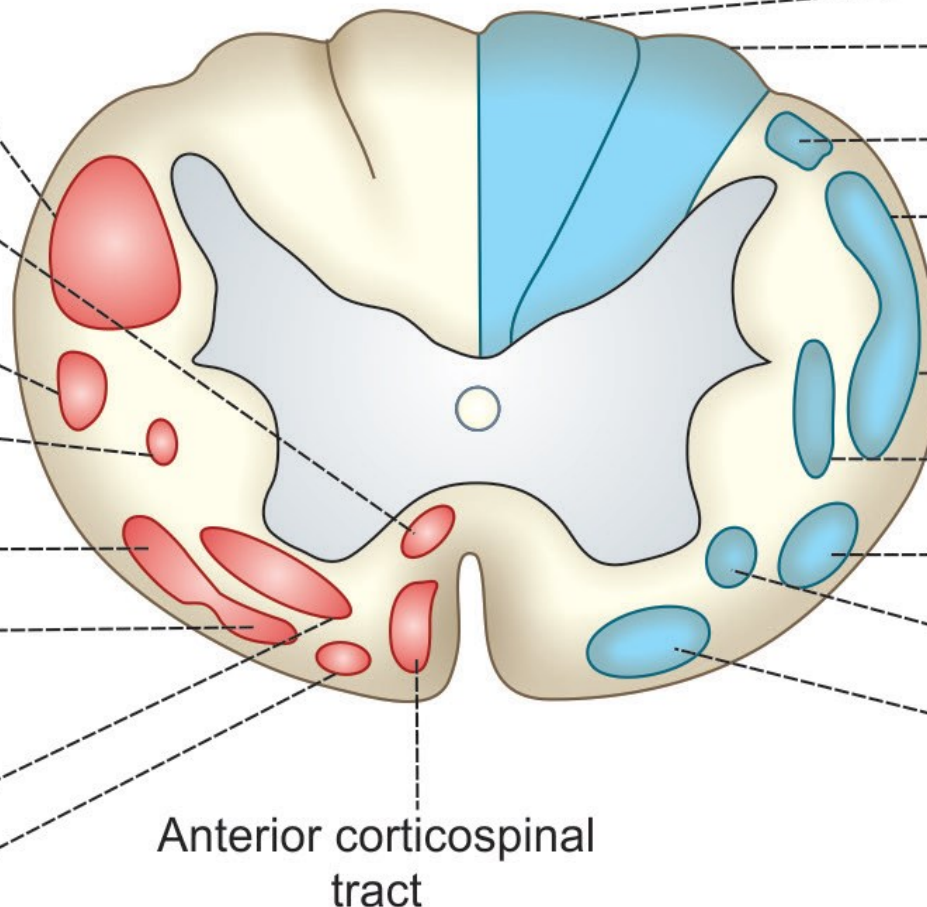


Descending tracts

Lateral corticospinal tract
Medial longitudinal fasciculus
Rubrospinal tract
Olivospinal tract
Lateral vestibulospinal tract
Anterior vestibulospinal tract
Reticulospinal tract
Tectospinal tract
Anterior corticospinal tract

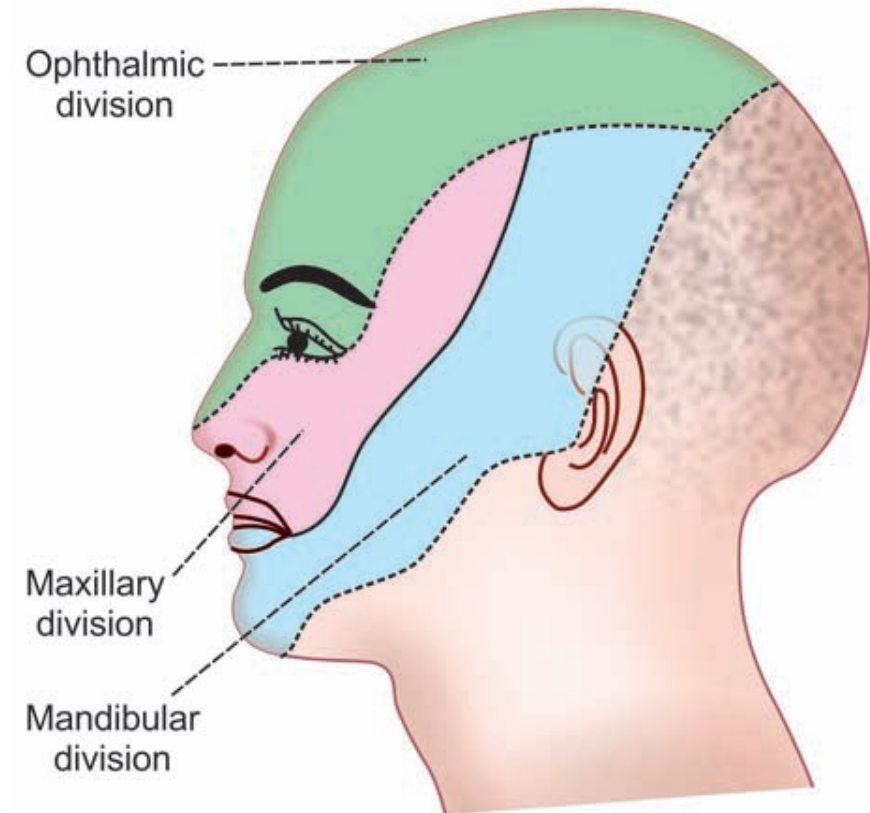
Ascending tracts

Fasciculus gracilis
Fasciculus cuneatus
Fasciculus dorsolateralis
Dorsal spinocerebellar tract
Ventral spinocerebellar tract
Lateral spinocerebellar tract
Spino-olivary tract
Spinotectal tract
Anterior spinothalamic tract



Somatic sensation from the head

- Cutaneous field of the head/face
 - Trigeminal nerve/pathway - from face
 - Ophthalmic branch – upper face
 - Maxillary “ – middle face/upper teeth...
 - Mandibular “ – lower face/also mastication (sensory and motor)
 - Cervical nerves (from back of the head)
 - C-2,3,4



	Dorsa Column Pathways	Anterolateral Pathways
Type	Fine discriminative touch, pressure, Vibration, proprioception/joint receptors	Crude touch, pressure, Pain, Temperature, Tickle, Itch, Sexual sensation
Range	Discrete	Wide (poor discrimination)
Types of fibers	Thick mylinated (A-alpha + A-beta)	Thin mylinated (A δ + C) or unmyelinated
Speed of transmission	Fast (30-110 ms)	Slow (0.5-20 ms)
1 st order neuron	Long- ipsilateral-- in spinal cord - synapse in medulla	Short-ends in dorsal horn spinal cord
2 nd order neuron	Crossed & to thalamus	Crosses (Dorsal spinal root) ascend to thalamus
3 rd order neuron	Thalamus to sensory cortex	Thalamus to cortex

The trigeminal system

Termination of 1st order neuron @

1. Mesencephalic nucleus (ASC)

*Proprioception
vibration*

2. Main sensory nuclei (ASC, mid pons)

For fine touch (Join med lemns)

3. Spinal trigeminal laminscus (3, DESC)

~ spinothalamic

For pain

Temperature

1st order – trigeminal ganglia

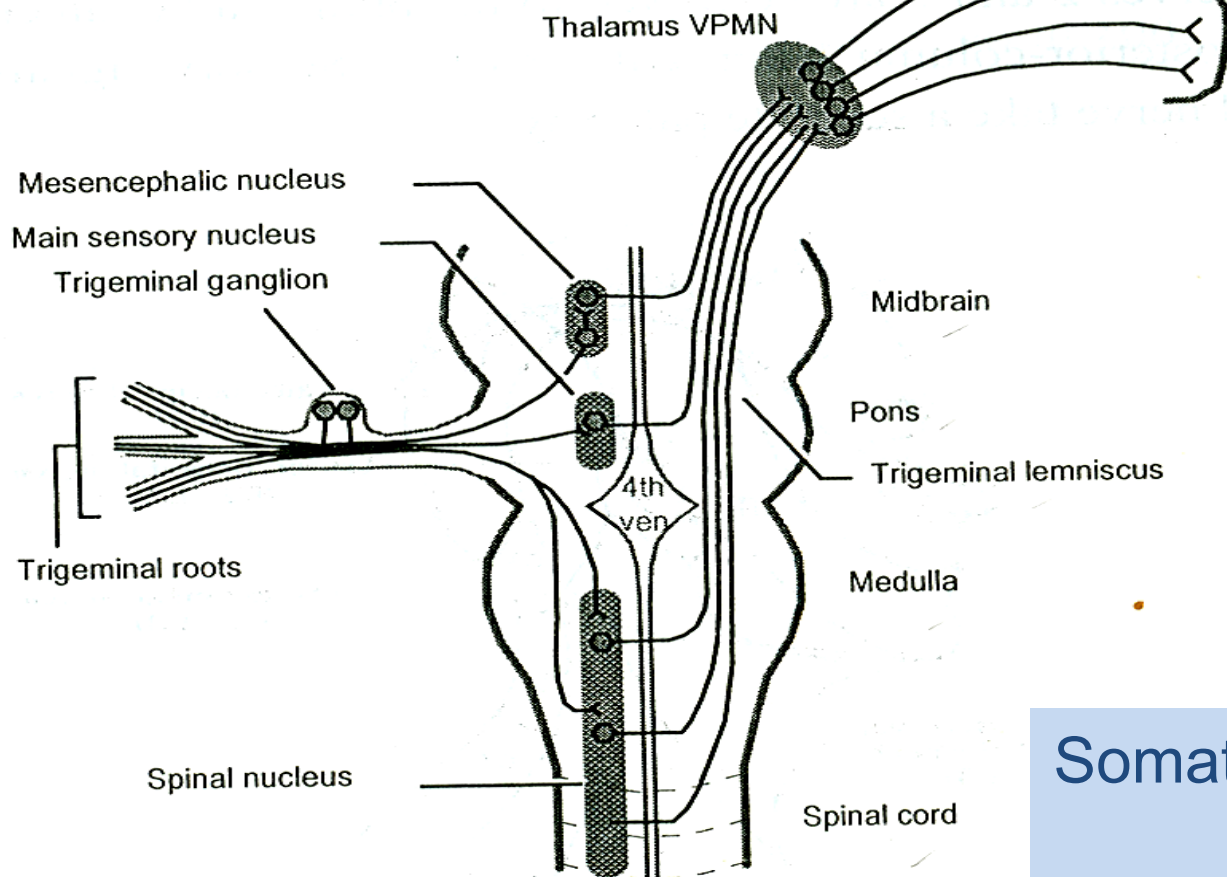
2nd order – from mesencephalic +

sensory + spinal nuclei and *cross & ascend* as trigeminal leminiscus

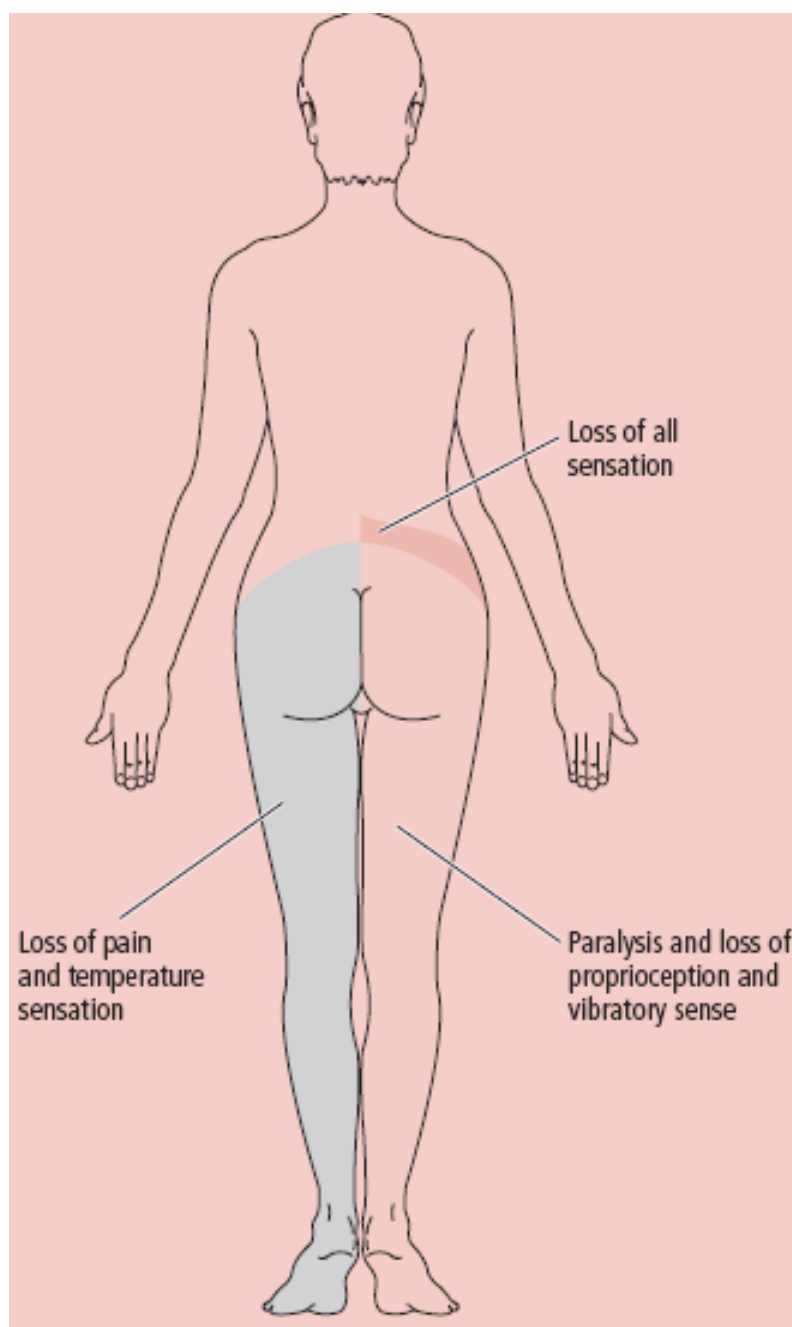
3rd order neurons = *VPMNT*

4th order neurons =

*final sensory
cerebral cortex
(inferior part)*



Somatic sensation from the head



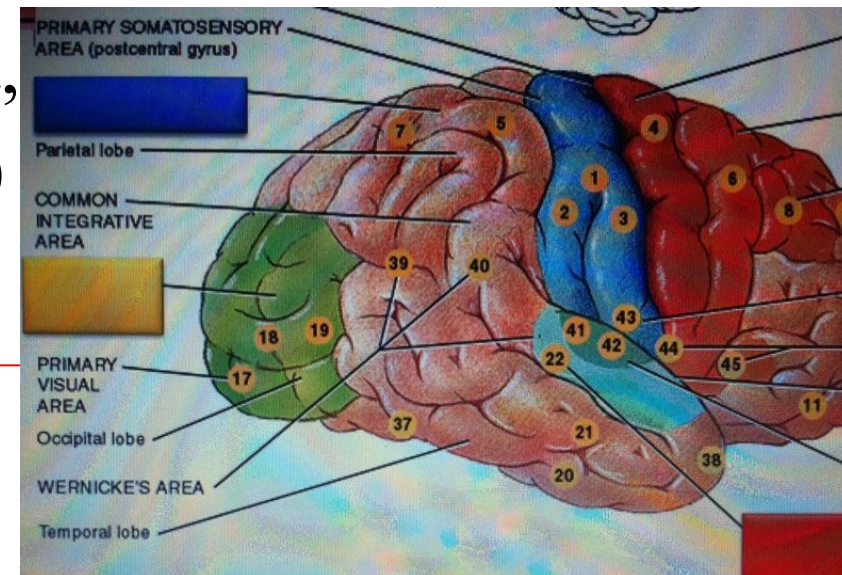
- ❖ A functional hemisection of the spinal cord
- ❖ Damage occurs to ipsilateral dorsal-column pathway, contralateral spinothalamic tract and ipsilateral descending motor (corticospinal tract) pathways.
- ❖ Ipsilateral loss of discriminative touch, vibration, and proprioception below the level of lesion.
- ❖ Contralateral loss of pain and temperature sensation beginning one or two segments below the lesion.
- ❖ Weakness and spasticity in certain muscle groups on the same side of the body.

Brown-Séquard syndrome

Although spinal cord injuries are rarely limited to a particular tract, quadrant, or side of the spinal cord, the hemisection of the spinal cord is used for instructive purposes. One such example is the **Brown-Séquard syndrome**

Sensory Functions of the Cerebral Cortex

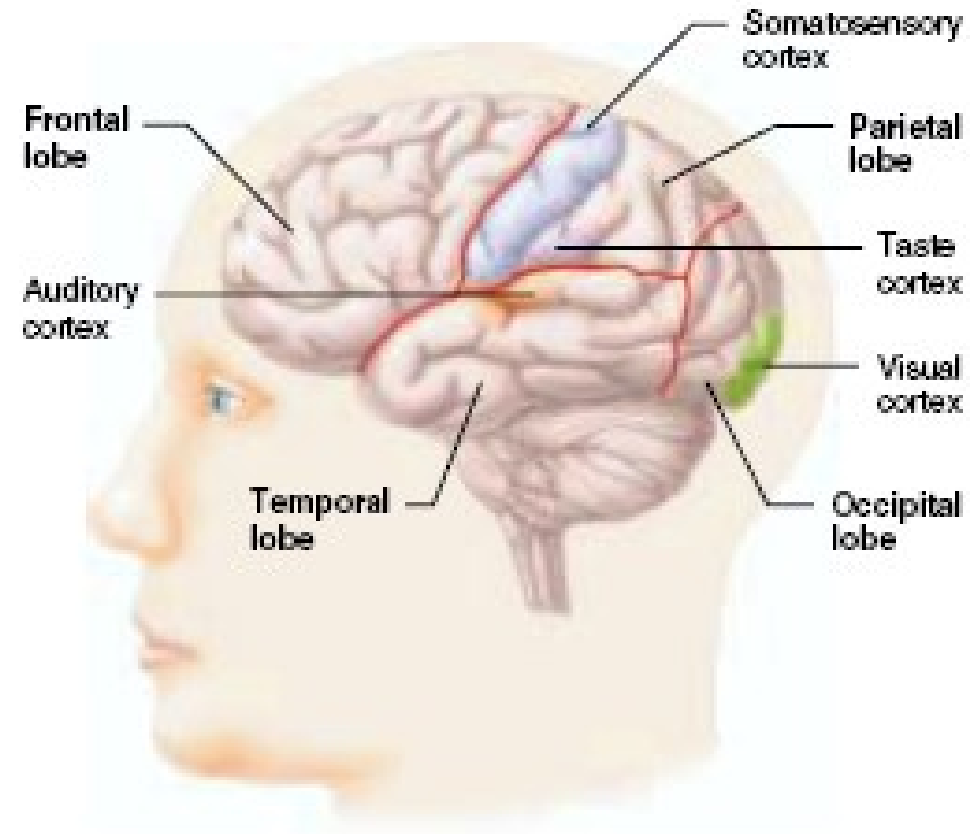
- **Somatosensory cerebral cortex:**
 - Highest center for conscious perception of **fine sensation**, whereas thalamic association nuclei perceive *crude touch, slow pain, and temperature*
- **Components sensory area of the cerebral cortex**
 - Primary somatic sensory area (SI, 1-2-3a,
 - Secondary somatic sensory area (SII, 40)
 - Somatic sensory association area (5,7)



- The cortex tell **what and where** answer, but not thalamus (only the presence of the modalities= *crude information*)

Functions of primary somatic sensory cortex = SI

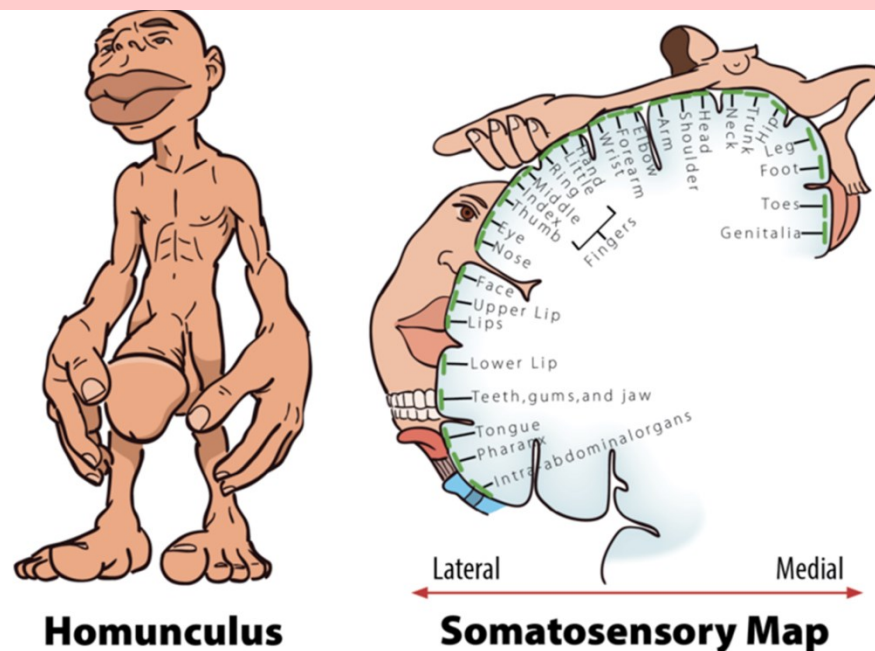
- Essential for normal conscious sensation
 - Perceptions of **fine touch**
 - Localization of **pain and temperature**
 - Intensity discrimination
 - Identification of texture
 - Proprioception
 - Inputs projection fibers from VPLNT + VPMNT
 - Send signals to **SS-II and SSA area**



Sensory Homunculus

- The representation in the primary sensory cortex
 - *Inverted* - the lower part of the body at the top of the cortex
 - *Crossed* – each half of the body on contralateral hemisphere.
 - *Unequal body mass* .
 - *Proportional to the density of the receptor in that specific area*
- proportional to the sensitivity of the body parts
(the more sensitive the area is the more area it takes).

Primary somatosensory cortex (S1)



Somato sensory association area (somatic interpretative area)



- Occupies area 5 and 7 = posterior to SI and SII.
- Receives signal from SS-I and SS-II
- *Combine all signals and gives meaning to complex sensations*
- The center of stereognosis
 - Agnosia/Astreognosis - inability to recognize/judge object through simple sensory skills/shapes or forms.
- Defines spatial orientation of body in surroundings
 - neglect syndrome/inattention/ignore contralateral side
 - Memory store of previous sensory experiences

CORTICAL AREA	FUNCTION
Sensory Input ↓	Relayed from afferent neuronal receptors
Primary Sensory Areas ↓	Initial cortical processing of sensory input
Unimodal Association Areas ↓	Further processing of information from a single sensory modality
Multimodal Sensory Association Areas ↓	Highest level of processing, integration, and interpretation of diverse sensory input for planning purposeful action
Multimodal Motor Association Areas ↓	Neuronal programming of movements according to cortical and subcortical input
Primary Motor Cortex	Transmission of impulses to somatic efferent motor neurons in spinal cord to initiate voluntary contraction of skeletal muscle